

Terms Rewrite Systems Equational Reasoning, Complete Property

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Abstract. Term rewriting systems (TRSs) provide a computational model for equational reasoning where equations are oriented into directed rules and applied as local transformations. A central question in this setting is whether a TRS is complete with respect to an underlying equational theory that is, whether rewriting alone is sufficient to decide equational validity. This paper surveys the main notions and methods surrounding this property. We first recall the standard framework of TRSs and equational theories, then examine the conditions under which completeness holds, with particular attention to how it can be preserved under extensions or modifications of the system. We review classical results connecting completeness to confluence and termination, discuss criteria for detecting its failure, and present the principal techniques for restoring it including Knuth-Bendix completion. This survey aims to give a unified and accessible account of completeness in term rewriting, serving as a reference for readers interested in the foundations of equational logic and automated deduction.

Keywords. term rewriting, confluence, termination, normalisation, complete, reasoning equational.

1 Introduction

Equational reasoning constitutes one of the foundational pillars of mathematical logic and theoretical computer science, providing a uniform framework for describing the algebraic properties of computational structures. Term Rewriting Systems (TRS) offer a particularly elegant and powerful approach to this problem, by orienting equational axioms into directed rewrite rules, thereby transforming the undirected question of equational provability into a deterministic computation of normal forms. Central to this framework is the notion of confluence the property ensuring that any two reduction sequences issuing from a common term eventually converge to a single result. Far from being a mere technical convenience, confluence is the cornerstone upon which the entire decision-theoretic apparatus of equational reasoning rests: It guarantees the uniqueness of normal forms, ensures the coherence of the induced equational theory, and, when combined with termination, yields the decidability of the word problem. In this paper, we investigate the role of confluence in the foundation of equational reasoning via TRS, examining its characterisations, its proof techniques no-

tably through Newman's Lemma and the Critical Pair Theorem and its consequences for the canonicity and completeness of rewriting-based decision procedures.

2 Preliminaries

2.1 Terms and signatures

Definition (*signature*). A *signature* Σ is a set of function symbols, each equipped with an *arity* $\text{ar} : \Sigma \rightarrow \mathbb{N}$. Symbols of arity 0 are called *constants*.

Nota. We fix two denumerable sets $\mathcal{X}$ and $\mathcal{C}$, of *variables* and *constants* respectively, disjoint from Σ and from each other.

Definition (*Terms*). A *term* over Σ and a set of atoms $\mathcal{A}$ (either $\mathcal{X}$ or $\mathcal{C}$) is defined inductively: every element $a \in \mathcal{A}$ is a term, and if $f \in \Sigma$ with $\text{ar}(f) = n$ and $t_1, \dots, t_n$ are terms, then $f(t_1, \dots, t_n)$ is a term.

Note. We write $\mathcal{T}(\Sigma, \mathcal{X})$ for the set of *patterns* (terms built over the variables $\mathcal{X}$), and $\mathcal{T}(\Sigma, \mathcal{C})$ for the set of *ground terms* (terms built over the constants $\mathcal{C}$, containing no variable).

Example. Let $\Sigma = \{0, s, +\}$ with arities 0, 1, 2 respectively (0 a constant, s a unary successor symbol, $+$ a binary addition symbol). Let $\mathcal{C} = \{a, b, c, \dots\}$ and $\mathcal{X} = \{x, y, z, \dots\}$.

- *Ground terms* ($\mathcal{T}(\Sigma, \mathcal{C})$, no variable): 0 , $s(0)$, $s(s(0))$, $+(s(0), s(s(0)))$, and also $+(a, s(b))$ once $a, b \in \mathcal{C}$ are used as free constants.
- *Patterns* ($\mathcal{T}(\Sigma, \mathcal{X})$, possibly containing variables): $+(x, s(0))$, $s(s(y))$, $+(x, +(y, z))$.

Ground terms are the terms with no variable occurrence; patterns are the general syntactic objects manipulated by rewrite rules, whose variables get instantiated by substitution.

Definition (*Substitution*). A *substitution* σ is a map $\sigma : \mathcal{X} \rightarrow \mathcal{T}(\Sigma, \mathcal{X})$.

Note. Generally, we want σ to have finite support, i.e.

$$|\{x \in \mathcal{X} : \sigma(x) \neq x\}| < +\infty.$$

Note (extension). A substitution does not apply only to variables, but to all terms, by structural extension:

- if x is a variable: $x\sigma = \sigma(x)$;
- if $f(t_1, \dots, t_n)$ is a term: $f(t_1, \dots, t_n)\sigma = f(t_1\sigma, \dots, t_n\sigma)$.

2.2 Equational theories

Definition (*Congruence*). Let A be a set equipped with a family of operations $(\omega_i)_{i \in I}$, each $\omega_i : A^{n_i} \rightarrow A$ of arity n_i (this is called an *algebra*, or Ω -*algebra*, in the sense of universal algebra; groups, rings, and term algebras are all instances of this notion, each with their own choice of operations ω_i for a group, $\Omega = \{\cdot, {}^{-1}, e\}$; for a term algebra over Σ , $\Omega = \Sigma$ itself, each symbol $f \in \Sigma$ acting as an operation of arity $\text{ar}(f)$). A *congruence*

on A is an equivalence relation $\sim$ on A that is *compatible* with every operation: for every $i \in I$ and every $a_1, \dots, a_{n_i}, b_1, \dots, b_{n_i} \in A$,

$$a_1 \sim b_1, \dots, a_{n_i} \sim b_{n_i} \implies \omega_i(a_1, \dots, a_{n_i}) \sim \omega_i(b_1, \dots, b_{n_i}).$$

Definition (*Quotient set induced by a congruence*). Given a congruence $\sim$ on A , the *quotient set* $A/\sim$ is the set of equivalence classes $[a]_\sim = \{b \in A : a \sim b\}$ for $a \in A$. Each operation ω_i descends to a well-defined operation on $A/\sim$, by

$$\omega_i([a_1]_\sim, \dots, [a_{n_i}]_\sim) := [\omega_i(a_1, \dots, a_{n_i})]_\sim,$$

this definition being independent of the choice of representatives $a_1, \dots, a_{n_i}$ precisely because $\sim$ is a congruence (this is exactly what compatibility guarantees).

Note. Specializing to $A = \mathcal{T}(\Sigma, \mathcal{X})$ with operations given by Σ itself (each $f \in \Sigma$ acting as an operation of arity $\text{ar}(f)$), a congruence on terms is exactly an equivalence relation $\sim$ such that $t_i \sim v_i$ for every i implies $f(t_1, \dots, t_n) \sim f(v_1, \dots, v_n)$ for every $f \in \Sigma$. This is the notion of congruence used throughout this paper.

Definition (*Equational theory*). An *equational theory* E over a signature Σ is a set of equations $s \approx t$ with $s, t \in \mathcal{T}(\Sigma, \mathcal{X})$.

Definition (*Equational congruence induced by E*). Given an equational theory E , we write $=_E$ for the smallest congruence on $\mathcal{T}(\Sigma, \mathcal{X})$ (Note above), stable under substitution, containing every instance $s\sigma \approx t\sigma$ for $s \approx t \in E$ and every substitution σ .

Note. Every equational theory E can be read as a term rewriting system by orienting each equation $s \approx t$ into a rule $s \rightarrow t$ or $t \rightarrow s$ this is precisely the purpose of term rewriting, introduced in the next section: to turn the undirected relation $=_E$ into a computable, directed process.

2.3 Rewrite rules

Definition (*Term Rewriting System (TRS)*). A *TRS* is a set R of rules $l \rightarrow_R r$ where $l, r \in \mathcal{T}(\Sigma, \mathcal{X})$, and we say that l *rewrites to* r . We also write (Σ, R) to say a set of rules on the alphabet or signature Σ .

Definition (*Iterated rewriting*). Let R be a TRS and $a, b \in \mathcal{T}(\Sigma, \mathcal{X})$. For $n \in \mathbb{N}$, we define $a \rightarrow_R^n b$ ("a rewrites to b in exactly n steps") by induction on n :

$$a \rightarrow_R^0 b \iff a = b, \quad a \rightarrow_R^{n+1} b \iff \exists c, a \rightarrow_R c \text{ and } c \rightarrow_R^n b.$$

We further write

$$a \rightarrow_R^* b \iff \exists n \in \mathbb{N}, a \rightarrow_R^n b, \quad a \rightarrow_R^+ b \iff \exists n \in \mathbb{N}^*, a \rightarrow_R^n b.$$

Definition (*Conversion*). Let R be a TRS. We write $a \leftrightarrow_R b$ if $a \rightarrow_R b$ or $b \rightarrow_R a$. For $n \in \mathbb{N}$, $a \leftrightarrow_R^n b$ is defined by induction exactly as in definition of iterated rewriting, with

$\leftrightarrow_R$ in place of $\rightarrow_R$:

$$a \leftrightarrow_R^0 b \iff a = b, \quad a \leftrightarrow_R^{n+1} b \iff \exists c, a \leftrightarrow_R c \text{ and } c \leftrightarrow_R^n b.$$

We write $a \leftrightarrow_R^* b$ if $\exists n, a \leftrightarrow_R^n b$: this is the smallest equivalence relation containing $\rightarrow_R$, called *conversion*.

Note. $\leftrightarrow_R^*$ is exactly the smallest congruence containing $\rightarrow_R$ and stable under substitution and context — it coincides with $=_E$ (definition of equational congruence induced by E) when R is obtained by orienting every equation of E (in either direction).

Definition (*Joinable*). Two terms a and b are joinable if exists a term c such that $a \rightarrow_R^* c$ and $b \rightarrow_R^* c$ we write it $c \downarrow_R^* b$ (assuming that R is terminating)

Definition (*Pre-diamond*). If a, b, c are three terms such that $b \leftarrow a \rightarrow c$, the triple (a, b, c) is called a *pre-diamond*, and a is its *root*. A pre-diamond is *proper* if a, b, c are pairwise distinct. A pre-diamond (a, b, c) is *joinable* if the pair (b, c) is joinable.

Note. A non-proper pre-diamond is always joinable.

Definition (*normal form*). Let R be a term rewriting system over $\mathcal{T}(\Sigma, \mathcal{X})$. A term t is said to be a *normal form* (or *irreducible*) for R if there is no term t' such that

$$t \rightarrow_R t'.$$

Definition (*Term with a normal form*). Let R be a TRS and $u \in \mathcal{T}(\Sigma, \mathcal{X})$. We say that u has a *normal form* if there exists a normal form $t \in \mathcal{T}(\Sigma, \mathcal{X})$ for R such that

$$u \leftrightarrow_R^* t.$$

Definition (*Weak normalization*). A TRS R is *weakly normalizing* (WN) if every term $t \in \mathcal{T}(\Sigma, \mathcal{X})$ has a normal form (definition of Term with a normal form), i.e.

$$\forall t \in \mathcal{T}(\Sigma, \mathcal{X}), \exists \bar{t} \in \mathcal{T}(\Sigma, \mathcal{X}) \text{ a normal form for } R \text{ such that } t \leftrightarrow_R^* \bar{t}.$$

Definition (*Position and subterm at a position*). Let $t \in \mathcal{T}(\Sigma, \mathcal{X})$. We define simultaneously, by induction on the structure of t , the set $\text{Pos}(t) \subseteq (\mathbb{N}_{>0})^* \cup \{\varepsilon\}$ of *positions* of t (finite sequences of positive integers), and, for every $p \in \text{Pos}(t)$, the *subterm of t at position p* , written $t|_p \in \mathcal{T}(\Sigma, \mathcal{X})$:

- If $t \in \mathcal{X} \cup \mathcal{C}$ (a variable or a constant):

$$\text{Pos}(t) = \{\varepsilon\}, \quad t|_\varepsilon = t.$$

- If $t = f(t_1, \dots, t_n)$ with $n \geq 1$:

$$\text{Pos}(t) = \{\varepsilon\} \cup \{i \cdot p : 1 \leq i \leq n, p \in \text{Pos}(t_i)\}, \quad t|_\varepsilon = t, \quad t|_{i \cdot p} = t_i|_p.$$

The empty sequence ε always denotes the *root* of t , with $t|_\varepsilon = t$.

Example. Let $t = f(g(x, y), z)$. Then

$$\text{Pos}(t) = \{\varepsilon, 1, 2, 1.1, 1.2\},$$

with

$$t|_{\varepsilon} = f(g(x, y), z), \quad t|_1 = g(x, y), \quad t|_2 = z, \quad t|_{1.1} = x, \quad t|_{1.2} = y.$$

Position 1 designates the first argument of f (here $g(x, y)$); positions starting with 1 then continue into the arguments of g (hence 1.1 and 1.2); position 2 designates the second argument of f (here z , a variable, hence with no further extension).

Definition (*Replacement at a position*). Let $t \in \mathcal{T}(\Sigma, \mathcal{X})$, $p \in \text{Pos}(t)$, and $r \in \mathcal{T}(\Sigma, \mathcal{X})$. The term $t[r]_p$ (t with its subterm at position p replaced by r) is defined by induction on p :

$$t[r]_{\varepsilon} = r, \quad f(t_1, \dots, t_n)[r]_{i.p} = f(t_1, \dots, t_i[r]_p, \dots, t_n).$$

By construction, $(t[r]_p)|_p = r$: replacing at p and then reading back at p recovers exactly r .

Example. With $t = f(g(x, y), z)$ as before, and $r = h(0)$:

$$t[r]_1 = f(h(0), z), \quad t[r]_2 = f(g(x, y), h(0)), \quad t[r]_{1.1} = f(g(h(0), y), z).$$

Nota.

- For $t \in \mathcal{T}(\Sigma, \mathcal{X})$ not a variable, $\text{root}(t)$ denotes the outermost function symbol of t : if $t = f(t_1, \dots, t_n)$, then $\text{root}(t) = f$.
- For p, q two positions, we write $p \leq q$ if p is a *prefix* of q , i.e. there exists p' such that $q = p \cdot p'$; in this case $t|_q = (t|_p)|_{p'}$.
- We write $p < q$ if $p \leq q$ and $p \neq q$ (*strict prefix*): the subterm at q occurs strictly inside the subterm at p .
- We write $p \parallel q$ if p and q are *disjoint* (or *incomparable*), i.e. neither $p \leq q$ nor $q \leq p$. In this case, for any term t with $p, q \in \text{Pos}(t)$ and any term r , we have $(t[r]_p)|_q = t|_q$ (definition of replacement at a position): replacing at p does not affect a disjoint position q .
- For two positions $p, q \in (\mathbb{N}_{>0})^*$, exactly one of the following holds:

$$p = q, \quad p < q, \quad q < p, \quad p \parallel q.$$

Definition (*Redex*). Let R be a TRS, t a term, and p a position in t . We say $t|_p$ is a *redex* if there exist $\ell \rightarrow r \in R$ and a substitution σ such that $t|_p = \ell\sigma$.

Definition (*Rewrite step*). Let R be a TRS, $t \in \mathcal{T}(\Sigma, \mathcal{X})$, $p \in \text{Pos}(t)$ such that $t|_p$ is a redex via rule $\ell \rightarrow r \in R$ and substitution σ (i.e. $t|_p = \ell\sigma$). The *rewrite step at position p using $\ell \rightarrow r$* replaces $t|_p$ by $r\sigma$ inside t :

$$t \xrightarrow[p]{\ell \rightarrow r} t[r\sigma]_p.$$

We write $t \rightarrow_R t'$ if $t' = t[r\sigma]_p$ for some rule $\ell \rightarrow r \in R$, position $p \in \text{Pos}(t)$, and substitution σ with $t|_p = \ell\sigma$: this is the relation already used in every definition above.

Example. Let $\Sigma = \{0, s, +\}$ and $R = \{+(0, y) \rightarrow y, +(s(x), y) \rightarrow s(+ (x, y))\}$. Consider $t = +(s(0), s(0))$.

- $\text{Pos}(t) = \{\varepsilon, 1, 2, 1 \cdot 1, 2 \cdot 1\}$ (writing 1, 2 for the two arguments of $+$, and $1 \cdot 1, 2 \cdot 1$ for the argument of each s).
- At $p = \varepsilon$: $t|_\varepsilon = t = +(s(0), s(0))$ is a redex via the second rule, with $\sigma(x) = 0$, $\sigma(y) = s(0)$. So

$$t \xrightarrow[\varepsilon]{+(s(x), y) \rightarrow s(+ (x, y))} s(+ (0, s(0))).$$

- In the resulting term $s(+ (0, s(0)))$, the subterm at position 1 is $+ (0, s(0))$, a redex via the first rule with $\sigma(y) = s(0)$:

$$s(+ (0, s(0))) \xrightarrow[1]{+(0, y) \rightarrow y} s(s(0)).$$

So $t \rightarrow_R^+ s(s(0))$ (in two steps), and $s(s(0))$ is a normal form: it has no redex, since no subterm of $s(s(0))$ matches a left-hand side of R .

Definition (Unifier). A substitution σ is a *unifier* of two terms s and t if $s\sigma = t\sigma$. A *most general unifier* (MGU) is a unifier σ such that for every unifier τ there exists a substitution θ with $\tau = \sigma \circ \theta$.

3 Core Properties

Definition (Confluence). A TRS R is *confluent* if, for all terms $s, t, u \in \mathcal{T}(\Sigma, \mathcal{X})$ with $s \rightarrow_R^* t$ and $s \rightarrow_R^* u$, there exists v such that $t \rightarrow_R^* v$ and $u \rightarrow_R^* v$.

Remark. This definition can be read through the lens of §2.1: if R orients every equation of an equational theory E (see the note following the definition of equational congruence induced by E), confluence of R is precisely what guarantees that joinability by $\rightarrow_R^*$ coincides with (and thus effectively computes) equality in $=_E$ see the theorem on confluence and termination and the discussion of the word problem below.

Definition (Local confluence). A TRS R is *locally confluent* if for all $t, u, v \in \mathcal{T}(\Sigma, \mathcal{X})$ with $t \rightarrow_R u$ and $t \rightarrow_R v$, there exists h such that $u \rightarrow_R^* h$ and $v \rightarrow_R^* h$.

Equivalently, a TRS R is locally confluent iff every pre-diamond is joinable.

Definition (Church–Rosser property). A TRS R is called *Church–Rosser* if for all $a, b \in \mathcal{T}(\Sigma, \mathcal{X})$ such that $a \leftrightarrow_R^* b$ (definition of conversion), there exists v such that $a \rightarrow_R^* v$ and $b \rightarrow_R^* v$.

Definition (Termination). $\rightarrow$ is *terminating* if every rewrite sequence $t_0 \rightarrow_R t_1 \rightarrow_R t_2 \rightarrow_R \dots$ is finite.

Definition (*Noetherian order*). A strict ordering is *Noetherian* if every non-empty set has a minimal element

Definition (*Ordering relation associated with a TRS*). Given a terminating TRS R on $\mathcal{T}(\Sigma, \mathcal{X})$, we denote $\prec_R$ the relation:

$$\forall a, b \in \mathcal{T}(\Sigma, \mathcal{X}), \quad b \prec_R a \iff a \rightarrow_R^+ b.$$

Proposition (*Properties of $\prec_R$*).

1. Since $\rightarrow_R^+$ is the transitive closure of $\rightarrow_R$, it is trivial that $\prec_R$ is a pre-order.
2. Since R terminates, $\prec_R$ is furthermore antisymmetric, and thus a strict ordering relation.
3. Again since R terminates, $\prec_R$ is a Noetherian ordering.

Proof. 1. $\prec_R$ is transitive. Let $a, b, c \in \mathcal{T}(\Sigma, \mathcal{X})$ with $c \prec_R b$ and $b \prec_R a$, i.e. (definition of the ordering relation associated with a TRS) $b \rightarrow_R^+ c$ and $a \rightarrow_R^+ b$. Since $\rightarrow_R^+$ is the transitive closure of $\rightarrow_R$, $a \rightarrow_R^+ b$ and $b \rightarrow_R^+ c$ give $a \rightarrow_R^+ c$, i.e. $c \prec_R a$. Hence $\prec_R$ is transitive; it is a strict pre-order (irreflexivity being immediate, since R terminates and $a \rightarrow_R^+ a$ would give an infinite reduction $a \rightarrow_R a \rightarrow_R a \rightarrow_R \dots$).

2. $\prec_R$ is antisymmetric. Suppose, for contradiction, that $a \prec_R b$ and $b \prec_R a$ for some $a \neq b$. This means $b \rightarrow_R^+ a$ and $a \rightarrow_R^+ b$. Concatenating a witnessing sequence of steps for each, we obtain an infinite reduction $a \rightarrow_R \dots \rightarrow_R b \rightarrow_R \dots \rightarrow_R a \rightarrow_R \dots$, contradicting termination of R . Hence $\prec_R$ is antisymmetric, and therefore a strict ordering relation.

3. $\prec_R$ is Noetherian. Suppose, for contradiction, that some non-empty set $A \subseteq \mathcal{T}(\Sigma, \mathcal{X})$ has no minimal element for $\prec_R$. Fix $t_0 \in A$. Since t_0 is not minimal in A , there exists $t_1 \in A$ with $t_1 \prec_R t_0$. By induction, having built $t_0, \dots, t_n \in A$ with $t_n \prec_R \dots \prec_R t_0$, since t_n is not minimal in A there exists $t_{n+1} \in A$ with $t_{n+1} \prec_R t_n$. This produces an infinite sequence $t_0 \succ_R t_1 \succ_R t_2 \succ_R \dots$ in A . By definition of $\prec_R$, each $t_k \succ_R t_{k+1}$ means $t_k \rightarrow_R^+ t_{k+1}$, i.e. there exists a finite non-empty sequence of rewrite steps from t_k to t_{k+1} . Concatenating these sequences for every k yields an infinite reduction sequence issued from t_0 , contradicting termination of R . Hence every non-empty set has a $\prec_R$ -minimal element: $\prec_R$ is Noetherian. $\square$

Definition (*Complete*). R is *complete*, or *convergent*, if R is terminating and confluent.

Remark. If R is convergent and orients an equational theory E (i.e. $\leftrightarrow_R^* = =_E$, definition of equational congruence induced by E), then for all $t, t' \in \mathcal{T}(\Sigma, \mathcal{X})$:

$$t =_E t' \iff \exists u, t \rightarrow_R^* u \text{ and } t' \rightarrow_R^* u.$$

Indeed, $t =_E t'$ means $t \leftrightarrow_R^* t'$; since R is confluent, this conversion yields a common reduct u by joinability. This is precisely what makes the word problem for E decidable once R is convergent: compute the (unique) normal forms of t and t' and compare them syntactically.

Proposition (*Canonical representatives via convergence*). Let E be an equational theory and R a convergent TRS orienting E (i.e. $\leftrightarrow_R^* = =_E$). Then the map sending each equivalence class $[t]_{=E} \in \mathcal{T}(\Sigma, \mathcal{X})/_E$ to the (unique) normal form of any of its representatives

is well defined, and gives a bijection

$$\mathcal{T}(\Sigma, \mathcal{X}) / \equiv_E \xrightarrow{\sim} \{\text{normal forms of } R\}.$$

Proof. Well defined. Let $t, t' \in [u]_{\equiv_E}$, i.e. $t =_E t'$, i.e. $t \leftrightarrow_R^* t'$. Since R is confluent, t and t' have a common reduct w (definition of confluence); since R terminates, t and t' each reduce to some normal form, and by confluence these normal forms coincide (a normal form cannot reduce further, so the common reduct w itself must reduce to, and hence equal, the normal form of both t and t'). Hence the normal form does not depend on the choice of representative t within the class.

Injective. If two classes $[t]_{\equiv_E}, [t']_{\equiv_E}$ have the same normal form $\bar{u}$, then $t \rightarrow_R^* \bar{u} \leftarrow_R^* t'$, so $t \leftrightarrow_R^* t'$, i.e. $t =_E t'$: the two classes coincide.

Surjective. Every normal form $\bar{u}$ is trivially the normal form of its own class $[\bar{u}]_{\equiv_E}$. $\square$

Theorem (Newman's Lemma). A locally confluent and terminating term rewriting system is confluent.

Proof. Since R is terminating, the relation $\rightarrow_R$ is well-founded, and thus induces an order relation $\prec_R$:

where $a \rightarrow_R^+ b$ means that a rewrites to b after one or more steps.

The fact that R is terminating allows us to carry out a proof by induction to show the property P defined by

$$\forall a \in \mathcal{T}(\Sigma, \mathcal{X}), \quad P(a) : \forall b, c \in \mathcal{T}(\Sigma, \mathcal{X}) \text{ such that } a \rightarrow_R^* b \text{ and } a \rightarrow_R^* c, \exists d \in \mathcal{T}(\Sigma, \mathcal{X}), b \rightarrow_R^* d \text{ and } c \rightarrow_R^* d.$$

Base case. The base case concerns terms a that do not rewrite to any other term, i.e.

$$\nexists t \in \mathcal{T}(\Sigma, \mathcal{X}), t \neq a \text{ and } a \rightarrow_R^* t.$$

Let such an a be given. In this case, for $b, c \in \mathcal{T}(\Sigma, \mathcal{X})$ such that $a \rightarrow_R^* b$ and $a \rightarrow_R^* c$, we can only have $c = b = a$. So $d = a$ gives us $b \rightarrow_R^* d$ and $c \rightarrow_R^* d$. Hence $P(a)$.

Inductive case. Let $a \in \mathcal{T}(\Sigma, \mathcal{X})$; we assume that for all a' such that $a' \prec_R a$, $P(a')$ holds. Let us show $P(a)$.

Let $b, c \in \mathcal{T}(\Sigma, \mathcal{X})$ such that $a \rightarrow_R^* c$ and $a \rightarrow_R^* b$.

This can also be written $a \rightarrow_R c_1 \rightarrow_R^* c$ and $a \rightarrow_R b_1 \rightarrow_R^* b$, with $c_1, b_1 \in \mathcal{T}(\Sigma, \mathcal{X})$, since $a \rightarrow_R c_1$ and $a \rightarrow_R b_1$.

By local confluence, there exists $u \in \mathcal{T}(\Sigma, \mathcal{X})$ such that $c_1 \rightarrow_R^* u$ and $b_1 \rightarrow_R^* u$.

Let such a u be given. We have $c_1 \rightarrow_R^* u$ and $c_1 \rightarrow_R^* c$, and moreover $c_1 \prec_R a$, so $P(c_1)$ holds. Hence there exists $e \in \mathcal{T}(\Sigma, \mathcal{X})$ such that $u \rightarrow_R^* e$ and $c \rightarrow_R^* e$.

Let such an e be given. Now, since $b_1 \rightarrow_R^* b$ and $b_1 \rightarrow_R^* u$, and moreover $b_1 \prec_R a$, we have $P(b_1)$. So there exists $v \in \mathcal{T}(\Sigma, \mathcal{X})$ such that $b \rightarrow_R^* v$ and $u \rightarrow_R^* v$.

We now have $u \rightarrow_R^* e$ and $u \rightarrow_R^* v$, and moreover $u \prec_R a$, so $P(u)$ holds. Hence there exists $s \in \mathcal{T}(\Sigma, \mathcal{X})$ such that $e \rightarrow_R^* s$ and $v \rightarrow_R^* s$, which concludes the proof. $\square$

Proof. Let R be a terminating and locally confluent TRS. Let

$$A = \{a \in \mathcal{T}(\Sigma, \mathcal{X}) \mid \forall c, b \in \mathcal{T}(\Sigma, \mathcal{X}) \text{ such that } c \leftarrow_R^+ a \rightarrow_R^+ b, \nexists d \in \mathcal{T}(\Sigma, \mathcal{X})\}$$

We prove that A is empty by contradiction. Assume that $A \neq \emptyset$.

Let a be the minimal element of A . Such an element exists because the rewrite relation $\rightarrow_R$ is terminating (i.e., well-founded).

Let b and c be two terms such that

$$a \rightarrow_R^* b \quad \text{and} \quad a \rightarrow_R^* c.$$

If both reductions from a to b and c occur in a single rewriting step, then, by local confluence, there exists a term d such that

$$b \rightarrow_R^* d \quad \text{and} \quad c \rightarrow_R^* d.$$

Hence, b and c are joinable, contradicting the assumption that $a \in A$.

Therefore, at least one of the two reductions must have length greater than one. Without loss of generality, there exist terms b_1 and c_1 such that

$$b \leftarrow_R^* b_1 \leftarrow_R a \rightarrow_R c_1 \rightarrow_R^* c.$$

Since $a \rightarrow_R b_1$ and $a \rightarrow_R c_1$ are single-step reductions, local confluence yields a term e satisfying

$$b_1 \rightarrow_R^* e \quad \text{and} \quad c_1 \rightarrow_R^* e.$$

Because $c_1 \prec a$, the minimality of a implies that $c_1 \notin A$. Hence, c and e are joinable in some term u .

Similarly, since $b_1 \prec a$, we have $b_1 \notin A$, and therefore b and e are joinable in some term f .

Finally, since $e \prec a$, we also have $e \notin A$. Thus, the terms u and f are joinable in some term g .

It follows that b and c are joinable, contradicting the assumption that $a \in A$.

Therefore, A is empty. Consequently, every terminating and locally confluent rewrite system is confluent. $\square$

Theorem (*Confluence and termination*). A terminating TRS is confluent if and only if it is locally confluent.

Proof. The previous proof already establishes that a terminating and locally confluent TRS is confluent. Let us now prove the converse, namely that a terminating confluent TRS is locally confluent. This is immediate, since local confluence is a particular case of confluence. $\square$

Lemma. If $\rightarrow_R$ is confluent, and for all terms a, b such that $a \leftrightarrow_R^* b$ exists a term c such that $a \rightarrow_R^* c \leftarrow_R^* b$

Proof. Let P be the predicate on $\mathbb{N}$ defined by

$\forall n \in \mathbb{N}, \quad P(n) : \text{“For every conversion sequence } a \leftrightarrow_R^* b \text{ of length } n, \text{ there exists a term } c \text{ such that } a \rightarrow_R^* c \leftarrow_R^* b \text{”}$

Base case ($n = 0$). If

$$a \leftrightarrow_R^0 b,$$

then $a = b$. Choosing $c = a$, we immediately obtain

$$a \rightarrow_R^* c \leftarrow_R^* b.$$

Hence, $P(0)$ holds.

Induction step. Let $n \in \mathbb{N}$ and assume that $P(n)$ holds.

Consider a conversion sequence of length $n + 1$,

$$a_1 \leftrightarrow_R a_2 \leftrightarrow_R \cdots \leftrightarrow_R a_n \leftrightarrow_R a_{n+1}.$$

By the induction hypothesis, there exists a term e such that

$$a_1 \rightarrow_R^* e \quad \text{and} \quad a_n \rightarrow_R^* e.$$

We distinguish two cases.

- If

$$a_n \rightarrow_R a_{n+1},$$

then, by confluence, the terms e and a_{n+1} are joinable. Hence, there exists a term u such that

$$e \rightarrow_R^* u \quad \text{and} \quad a_{n+1} \rightarrow_R^* u.$$

Since $a_1 \rightarrow_R^* e$, it follows that

$$a_1 \rightarrow_R^* u \quad \text{and} \quad a_{n+1} \rightarrow_R^* u.$$

- If

$$a_{n+1} \rightarrow_R a_n,$$

then

$$a_{n+1} \rightarrow_R a_n \rightarrow_R^* e,$$

and therefore

$$a_{n+1} \rightarrow_R^* e.$$

Together with $a_1 \rightarrow_R^* e$, we conclude that

$$a_1 \rightarrow_R^* e \quad \text{and} \quad a_{n+1} \rightarrow_R^* e.$$

Thus, in both cases, there exists a term that is reachable from both a_1 and a_{n+1} . Therefore, $P(n + 1)$ holds.

By induction, $P(n)$ is true for every $n \in \mathbb{N}$, which completes the proof. $\square$

Theorem (*Termination and Church–Rosser*). A terminating TRS R is Church–Rosser if and only if it is locally confluent.

Proof. The previous lemma establishes the first implication,

$$\text{Confluence} \implies \text{Church–Rosser}.$$

We now prove the converse implication.

Assume that the rewrite relation $\rightarrow_R$ satisfies the Church–Rosser property. We show that $\rightarrow_R$ is confluent, that is,

$$\forall a, b, c, \quad a \rightarrow_R^* b \text{ and } a \rightarrow_R^* c \implies \exists d, \quad b \rightarrow_R^* d \text{ and } c \rightarrow_R^* d.$$

Let a , b , and c be terms such that

$$a \rightarrow_R^* b \quad \text{and} \quad a \rightarrow_R^* c.$$

Then,

$$c \leftarrow_R a \rightarrow_R^* b,$$

which implies

$$c \leftrightarrow_R^* a \leftrightarrow_R^* b.$$

Hence,

$$c \leftrightarrow_R^* b.$$

Since $\rightarrow_R$ satisfies the Church–Rosser property, there exists a term d such that

$$c \rightarrow_R^* d \quad \text{and} \quad b \rightarrow_R^* d.$$

Therefore, every pair of reductions starting from the same term can be joined, proving that $\rightarrow_R$ is confluent. □

4 Critical pairs

The property of confluence is very important for a TRS, especially for equational reasoning. It allows us to rewrite an expression in different ways and still obtain the same result. That is why it is important to have this property.

Generally, it is very difficult to obtain a confluent TRS, or to make a TRS confluent using classical methods. But in the special case of terminating TRSs, the previous equivalence reduces the problem to simply ensuring that the TRS is locally confluent in order to make it confluent.

In this context, the objective is: how can we make a TRS locally confluent?

Definition (Overlap). Let R be a term rewriting system. An *overlap* occurs when there exist rules $l_1 \rightarrow r_1$ and $l_2 \rightarrow r_2$ in R , a non-variable position $p \in \text{Pos}(l_2)$, and a substitution σ such that

$$l_2|_p \sigma = l_1 \sigma.$$

In other words, the left-hand side l_1 matches (up to substitution) a non-variable subterm of the left-hand side l_2 .

Definition (Critical pair). Let $l_1 \rightarrow_R r_1$ and $l_2 \rightarrow_R r_2$ be two rewrite rules sharing no variables (renamed apart if necessary). Suppose there exists a position $p \in \text{Pos}_{\mathcal{F}}(l_1)$ (i.e., p is *not* a variable position of l_1) and a most general unifier σ such that

$$l_1|_p \sigma = l_2 \sigma.$$

This gives rise to two possible reductions of $l_1 \sigma$:

1. $l_1 \sigma \rightarrow r_1 \sigma$;
2. $l_1 \sigma \rightarrow l_1 \sigma[r_2 \sigma]_p$.

We call *critical pair* the pair $(r_1 \sigma, l_1 \sigma[r_2 \sigma]_p)$. By convention, the trivial overlap of a rule with itself at the root position ($l_1 = l_2$, $p = \varepsilon$, $\sigma = \text{id}$) is excluded, as it always yields the trivial pair (r_1, r_1) .

Algorithm 1 ComputeCriticalPairs

Require: A term rewriting system $R = \{l_1 \rightarrow_R r_1, \dots, l_n \rightarrow_R r_n\}$

Ensure: The set $CP(R)$ of all (non-trivial) critical pairs of R

```
1: function COMPUTECRITICALPAIRS( $R$ )
2:    $CP \leftarrow \emptyset$ 
3:   for all rule  $(l_1 \rightarrow_R r_1) \in R$  do
4:     for all rule  $(l_2 \rightarrow_R r_2) \in R$  do
5:        $(l'_2, r'_2) \leftarrow \text{RENAME}(l_2 \rightarrow_R r_2)$   $\triangleright$  fresh variables, so  $l_1 \rightarrow_R r_1$  and  $l'_2 \rightarrow_R r'_2$ 
        share none
6:       for all position  $p \in \text{Pos}_{\mathcal{F}}(l_1)$  do
7:         if  $(l_1 \rightarrow_R r_1) = (l_2 \rightarrow_R r_2)$  and  $p = \varepsilon$  then
8:           continue  $\triangleright$  skip the trivial root self-overlap
9:         end if
10:         $\sigma \leftarrow \text{MGU}(l_1|_p, l'_2)$ 
11:        if  $\sigma \neq \text{FAIL}$  then
12:           $t_1 \leftarrow \sigma(r_1)$ 
13:           $t_2 \leftarrow \sigma(l'_2|_p)$ 
14:           $CP \leftarrow CP \cup \{(t_1, t_2)\}$ 
15:        end if
16:      end for
17:    end for
18:  end for
19:  return  $CP$ 
20: end function
```

Note. Critical pairs are therefore important for studying confluence, because they capture all the points of conflict.

Note. Let R be a term rewriting system, and let (s, t) be a critical pair of R . The critical pair is said to be *trivial* if $s \downarrow_R t$, that is, if s and t are joinable. In particular, every critical pair satisfying $s = t$ is trivial.

Definition (*Parallel rewriting*). Let R be a set of rewrite rules. For a term t , a parallel rewrite step is a relation

$$t \Rightarrow_R t'$$

defined as follows.

There exists a finite non-overlapping set of positions

$$P = \{p_1, \dots, p_n\} \subseteq \text{Pos}(t)$$

(i.e., no p_i is a prefix of another position in P), and for each $i \in \{1, \dots, n\}$ there exist a rule $\ell_i \rightarrow r_i \in R$, a most general unifier (matching substitution) σ_i and the condition

$$t|_{p_i} = \ell_i \sigma_i,$$

such that

$$t' = t[r_1 \sigma_1]_{p_1} \cdots [r_n \sigma_n]_{p_n}.$$

Equivalently: at each position $p_i \in P$ we simultaneously replace the matched subterm $t|_{p_i}$ by the corresponding instance $r_i \sigma_i$, while all positions not in P remain unchanged. (Optionally, one may require $P \neq \emptyset$ for a proper step.)

Note. $\rightarrow_R \subseteq \Rightarrow_R \subseteq \rightarrow_R^*$

Definition (*context*). Let Σ be a signature and $\mathcal{X}$ a set of variables. Let $\{\square_1, \dots, \square_n\}$ be a set of fresh symbols, called *holes*, disjoint from $\Sigma \cup \mathcal{X}$, each treated as a nullary symbol (a constant) for the purpose of forming terms. A *context* with holes $\square_1, \dots, \square_n$ is a term

$$C \in \mathcal{T}(\Sigma, \mathcal{X} \cup \{\square_1, \dots, \square_n\}).$$

Given terms $s_1, \dots, s_n \in \mathcal{T}(\Sigma, \mathcal{X})$, we write $C[s_1, \dots, s_n]$ for the term obtained from C by simultaneously replacing every occurrence of $\square_k$ by s_k , for each $k = 1, \dots, n$. Formally, if $q_1, \dots, q_m \in \text{Pos}(C)$ are the positions at which some hole occurs, with $C|_{q_j} = \square_{k_j}$, then

$$C[s_1, \dots, s_n] = C[s_{k_1}]_{q_1} [s_{k_2}]_{q_2} \cdots [s_{k_m}]_{q_m}.$$

Nota. Let C be a context with holes $\square_1, \dots, \square_n$. We say that C is *linear* in $\square_1, \dots, \square_n$ if each hole $\square_k$ occurs at exactly one position of C :

$$\forall k \in \{1, \dots, n\}, \quad |\{q \in \text{Pos}(C) \mid C|_q = \square_k\}| = 1.$$

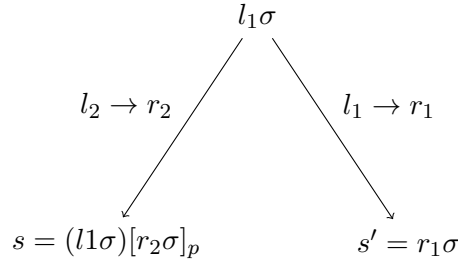
Equivalently, the replacement $C[s_1, \dots, s_n]$ inserts each s_k at a single, well-defined position of C , which we denote $p_k \in \text{Pos}(C)$.

Theorem (*Critical Pair Theorem (Knuth–Bendix)*). A terminating TRS is confluent if and only if both terms of each of its critical pairs reduce to the same term.

Proof. Let R be a terminating term rewriting system (TRS).

We first assume that R is confluent.

Let (s, s') be a critical pair of R . By definition, there exists two rules $l_1 \rightarrow r_1, l_2 \rightarrow r_2 \in R$, a most general unifier σ , and a position $p \in \mathcal{P}(l_1)$ such that $l_1|_p \sigma = l_2 \sigma$ with $s = (l_1 \sigma)[p \leftarrow r_2 \sigma]$ and $s' = r_1 \sigma$:



Since R is confluent, there exists a term u such that :

$$r_1 \sigma \rightarrow_R^* u \leftarrow_R^* l_1 \sigma [r_2 \sigma]_p$$

Thus all critical pairs are joinable.

Conversely, assume that every critical pair of R is joinable. We show that R is confluent.

Since R is terminating, Newman's Lemma implies that it is sufficient to prove that R is locally confluent. Therefore, it remains to show that every local peak

$$s \leftarrow_R t \rightarrow_R p$$

is joinable.

Let A be the set of terms a such that

$$\exists b, c \in \mathcal{T}(\Sigma, \mathcal{X}), \quad b \leftarrow_R a \rightarrow_R c, \quad \nexists d \in \mathcal{T}(\Sigma, \mathcal{X}), \quad b \rightarrow_R^* d \leftarrow_R^* c.$$

Assume that $A \neq \emptyset$.

Since the strict subterm ordering is well-founded. Therefore, there exists a minimal (for the subterm relation) element $a \in A$ such that:

$$\forall a' \in A, \quad a \neq a' \Rightarrow a \prec a',$$

where

$$x \prec y \iff y \rightarrow_R^+ x.$$

Let such a minimal term a be fixed. Let b, c be terms such that

$$b \leftarrow_R a \rightarrow_R c.$$

By definition of a rewrite step, there exist rewrite rules

$$l_1 \rightarrow_R r_1$$

and

$$l_2 \rightarrow_R r_2,$$

positions $p, q \in \text{Pos}(a)$ and substitutions σ_1, σ_2 such that

$$a|_p = l_1 \sigma_1, \quad a|_q = l_2 \sigma_2$$

and

$$b = a[r_1 \sigma_1]_p, \quad c = a[r_2 \sigma_2]_q.$$

We study the relative positions of p and q .

For two positions p and q : we say $p \leq q$ if p is a prefix of q , i.e. there exists q' such that $q = p \cdot q'$ (possibly $q' = \varepsilon$, in which case $p = q$); and p, q are *disjoint* if neither is a prefix of the other, in which case the two redexes occur in two independent subterms.

These cases are exhaustive because for two finite sequences of natural numbers, exactly one of the following holds:

$$p \leq q, \quad q \leq p, \quad \text{or } p, q \text{ disjoint.}$$

Case 1 (disjoint positions). Assume p and q are disjoint. The idea is that the two rewrite steps commute. In fact, replacing the subterm of a at position p does not modify any subterm at a position disjoint from p . Therefore,

$$(a[r_1 \sigma_1]_p)|_q = a|_q = l_2 \sigma_2.$$

Hence, after the first rewrite, the redex at position q is still present. We can apply the rule $l_2 \rightarrow r_2$:

$$a[r_1\sigma_1]_p \rightarrow a[r_1\sigma_1]_p[r_2\sigma_2]_q.$$

Symmetrically,

$$(a[r_2\sigma_2]_q)|_p = a|_p = l_1\sigma_1,$$

therefore

$$a[r_2\sigma_2]_q \rightarrow a[r_2\sigma_2]_q[r_1\sigma_1]_p.$$

Since p and q are disjoint, the two replacements act on two different subterms. Therefore they commute:

$$a[r_1\sigma_1]_p[r_2\sigma_2]_q = a[r_2\sigma_2]_q[r_1\sigma_1]_p.$$

Let this common term be d . Then $b \rightarrow d$ and $c \rightarrow d$. Hence the peak is joinable.

Case 2 (p and q comparable). Without loss of generality, assume $p \leq q$ (the case $q \leq p$ is symmetric, exchanging the roles of $(R_1, l_1, r_1, \sigma_1, p)$ and $(R_2, l_2, r_2, \sigma_2, q)$). Write $q = p \cdot q'$ with $q' \in \text{Pos}(l_1\sigma_1)$.

We distinguish two cases, according to whether q' lies below a variable of l_1 or not.

Case 2.1 (q' is not below a variable of l_1). Then $q' \in \text{Pos}(l_1)$ and $(l_1\sigma_1)|_{q'} = (l_1|_{q'})\sigma_1$.

Sub-case $q' = \varepsilon$ (same redex). Then $l_1\sigma_1 = l_2\sigma_2$: the two rules apply to the same redex $a|_p$. If the two rules are identical up to a renaming of variables (there exists a bijective substitution $\rho : \mathcal{X} \rightarrow \mathcal{X}$ with $l_2 = l_1\rho$, $r_2 = r_1\rho$), then $l_1\sigma_1 = l_1\rho\sigma_2$, and since variable names in rewrite rules are irrelevant, $r_1\sigma_1 = r_2\sigma_2$. Hence $b = a[r_1\sigma_1]_p = a[r_2\sigma_2]_p = c$, trivially joinable with $d = b = c$. Otherwise, unifying l_1 and l_2 yields a most general unifier μ and a critical pair $(r_1\mu, r_2\mu)$; by hypothesis it is joinable, giving d with $r_1\mu \rightarrow_R^* d$, $r_2\mu \rightarrow_R^* d$. Since $l_1\sigma_1 = l_2\sigma_2$ is an instance of this critical pair, there exists θ with $\sigma_1 = \mu\theta$ on $\text{Var}(l_1)$ and $\sigma_2 = \mu\theta$ on $\text{Var}(l_2)$; by stability under substitution, $r_1\sigma_1 = r_1\mu\theta \rightarrow_R^* d\theta$ and $r_2\sigma_2 = r_2\mu\theta \rightarrow_R^* d\theta$, so $b \rightarrow_R^* a[d\theta]_p$ and $c \rightarrow_R^* a[d\theta]_p$: the peak is joinable.

Sub-case $q' \neq \varepsilon$ (proper overlap). Since q' is not below a variable, q' is a non-variable position of l_1 : unifying $l_1|_{q'}$ and l_2 (variables of l_1, l_2 renamed apart) yields a most general unifier μ and a critical pair $(r_1\mu, l_1[r_2]_{q'}\mu)$; by hypothesis it is joinable, giving d with $r_1\mu \rightarrow_R^* d$ and $(l_1[r_2]_{q'})\mu \rightarrow_R^* d$. As above, there exists θ with $\sigma_1 = \mu\theta$ on $\text{Var}(l_1)$, $\sigma_2 = \mu\theta$ on $\text{Var}(l_2)$; by stability under substitution, $b = r_1\sigma_1 = r_1\mu\theta \rightarrow_R^* d\theta$, and $c = (l_1[r_2]_{q'})\sigma_1[\dots] = (l_1[r_2]_{q'})\mu\theta \rightarrow_R^* d\theta$ (using $l_1\sigma_1[r_2\sigma_2]_{q'} = a[r_2\sigma_2]_q = c$ via the position algebra); by stability under context, $b \rightarrow_R^* a[d\theta]_p$ and $c \rightarrow_R^* a[d\theta]_p$: the peak is joinable.

Case 2.2 (q' is below a variable of l_1). There exist $x \in \text{Var}(l_1)$, $p_x \in \text{Pos}(l_1)$ with $l_1|_{p_x} = x$, and $q'' \in \text{Pos}(x\sigma_1)$ such that $q' = p_x \cdot q''$ and $(x\sigma_1)|_{q''} = l_2\sigma_2$. Define $\sigma'_1(x) = (x\sigma_1)[r_2\sigma_2]_{q''}$, $\sigma'_1(y) = \sigma_1(y)$ for $y \neq x$. Since $q = p \cdot p_x \cdot q''$, replacing a at position q amounts to first replacing $a|_p$ at position $p_x \cdot q''$ and then reinserting at p :

$$c = a[r_2\sigma_2]_q = a[(a|_p)[r_2\sigma_2]_{p_x \cdot q''}]_p = a[l_1\sigma'_1]_p.$$

Since $l_1|_{p_x} = x$ is a variable, $l_1\sigma'_1$ still matches l_1 at the root, so we may apply $l_1 \rightarrow r_1$ at position p :

$$c = a[l_1\sigma'_1]_p \rightarrow a[r_1\sigma'_1]_p =: w.$$

The substitutions σ_1, σ'_1 differ only on x , and $x\sigma_1 \rightarrow_R x\sigma'_1$ is precisely the step $l_2\sigma_2 \rightarrow r_2\sigma_2$ at position q'' . If x occurs $k \geq 0$ times in r_1 , rewriting each occurrence of $x\sigma_1$ inside $r_1\sigma_1$ in turn gives $r_1\sigma_1 \rightarrow_R^* r_1\sigma'_1$ (trivially if $k = 0$), and by stability under context, $b = a[r_1\sigma_1]_p \rightarrow_R^* a[r_1\sigma'_1]_p = w$. Hence $b \rightarrow_R^* w$ and $c \rightarrow w$: the peak is joinable.

All local peaks are therefore joinable. Hence R is locally confluent. Since R is terminating, Newman's lemma implies that R is confluent. □

Theorem (*Union of confluent TRS*). The union of two confluent TRS is confluent if they don't share any function (or constants).

Note. The advantage of having distinguished signatures is that in both systems no rule of one system rewrites one term to the other. On the other hand, the union means that we can have mixed terms. To see the confluence we would like to see the application of the exit rules applied the confluence of each system

Definition (*pure term*). Let $R_i = (\Sigma_i, \mathcal{R}_i)$.

A term t is said to be *pure* for R_i if all of its symbols belong to Σ_i .

Formally:

$$t \in T(\Sigma_i, X).$$

Definition (*Alien term*). Let $t \in \mathcal{T}(\Sigma, \mathcal{X})$ be a term whose root belongs to Σ_i (we say that t is *i-rooted*: $\text{root}(t) \in \Sigma_i$). A subterm $s = t|_p$ (for some position $p \in \text{Pos}(t)$, $p \neq \varepsilon$) is called an *alien* of t (for R_i) if $\text{root}(s) \in \Sigma_j$, $j \neq i$, while every proper prefix of p carries a symbol of Σ_i . Equivalently: s is a maximal subterm of t whose root does not belong to Σ_i .

No condition is imposed on the symbols occurring strictly below the root of s : an alien may itself contain further aliens (rooted back in Σ_i , or beyond).

Definition (*term preserver*). Let $R_1 = (\Sigma_1, \mathcal{R}_1)$ and $R_2 = (\Sigma_2, \mathcal{R}_2)$ be two rewrite systems with disjoint signatures, $\Sigma_1 \cap \Sigma_2 = \emptyset$, acting on terms over $\Sigma = \Sigma_1 \cup \Sigma_2$.

For a term t and a position $p \in \text{Pos}(t)$, we call p an *alien position* (with respect to R_1) if the symbol at position p in t belongs to Σ_2 , while the symbol at every position strictly above p (i.e. every proper prefix of p) belongs to Σ_1 . The subterm $t|_p$ is then called an *alien* of t .

A rewrite rule $\ell \rightarrow r \in \mathcal{R}_1$ is called a *term preserver* if it introduces no new symbol from Σ_2 , that is, every function symbol occurring in r belongs either to Σ_1 or to the variables of ℓ :

$$\mathcal{F}(r) \subseteq \Sigma_1 \cup \text{Var}(\ell).$$

Concretely, for a step $t \xrightarrow[p]{\ell \rightarrow r} t'$ with $t|_p = \ell\sigma$ and $t' = t[r\sigma]_p$: any alien of t located

outside the rewritten position p is left unchanged in t' , and any alien located inside $t|_p$ is only substituted for a variable of ℓ , hence reappears unmodified in t' (possibly copied several times or deleted, but never altered). In other words, R_1 may rearrange, duplicate, or erase the aliens coming from R_2 , but it can never fabricate a new one.

Definition (*collapsing rule*). Let $R_i = (\Sigma_i, \mathcal{R}_i)$. A rule $\ell \rightarrow_R r \in \mathcal{R}_i$ is called *collapsing* if $r \in \mathcal{X}$, i.e. the right-hand side is a single variable. A rewrite step

$$t \xrightarrow[p]{\ell \rightarrow_R r} t'$$

is called a *collapsing step* if the rule $\ell \rightarrow_R r$ used is collapsing.

Nota. Let $t \in \mathcal{T}(\Sigma, \mathcal{X})$. We say that t is *preserved* (for $R_1 \cup R_2$) if no reduction sequence issued from t contains a collapsing step, i.e. for every

$$t = t_0 \rightarrow t_1 \rightarrow t_2 \rightarrow \dots$$

(finite or infinite), every step $t_k \rightarrow t_{k+1}$ uses a non-collapsing rule.

Definition (*skeleton decomposition*). Let t be i -rooted with aliens $s_1, \dots, s_n$, occurring at the (pairwise disjoint) alien positions $p_1, \dots, p_n$. There exists a unique context

$$C \in \mathcal{T}(\Sigma_i, \mathcal{X} \cup \{\square_1, \dots, \square_n\}),$$

linear in the fresh variables $\square_1, \dots, \square_n$, such that

$$t = C[s_1, \dots, s_n],$$

where $C[s_1, \dots, s_n]$ denotes the term obtained by replacing $\square_k$ by s_k at position p_k , for every k . We call C the *skeleton* of t .

Lemma (1). Let

$$R_1 = (\Sigma_1, \mathcal{R}_1) \quad \text{and} \quad R_2 = (\Sigma_2, \mathcal{R}_2)$$

be two term rewriting systems such that

$$\Sigma_1 \cap \Sigma_2 = \emptyset.$$

Then the union $R_1 \cup R_2$ satisfies the following: for every local peak

$$t \xrightarrow{R_1} s \xrightarrow{R_2} u,$$

there exists a term w such that

$$t \xrightarrow{*}_{R_2} w \quad \text{and} \quad u \xrightarrow{*}_{R_1} w.$$

Proof. By definition of a rewrite step, there exist rules

$$l_1 \rightarrow_R r_1 \in \mathcal{R}_1, \quad l_2 \rightarrow_R r_2 \in \mathcal{R}_2,$$

positions $p, q \in \text{Pos}(s)$, and substitutions σ_1, σ_2 such that

$$s|_p = l_1\sigma_1, \quad s|_q = l_2\sigma_2,$$

and

$$t = s[r_1\sigma_1]_p, \quad u = s[r_2\sigma_2]_q.$$

We study the relative positions of p and q . For two positions, exactly one of the following holds: $p = q$, $p < q$, $q < p$, or p and q are disjoint.

Preliminary claim. Let $i \in \{1, 2\}$, $l \rightarrow_R r \in \mathcal{R}_i$, σ a substitution, and $q' \in \text{Pos}(l\sigma)$. We claim that exactly one of the following holds:

- (a) q' is *not below a variable* of l (i.e. no strict prefix $p_x \subsetneq q'$ satisfies $l|_{p_x} \in \mathcal{X}$); in this case $q' \in \text{Pos}(l)$ and

$$\text{root}((l\sigma)|_{q'}) = \text{root}(l|_{q'}) \in \Sigma_i;$$

- (b) q' is *below a variable* of l : there exist a (unique) variable $x \in \text{Var}(l)$, a position $p_x \in \text{Pos}(l)$ with $l|_{p_x} = x$, and $q'' \in \text{Pos}(x\sigma)$, such that

$$q' = p_x \cdot q'', \quad (l\sigma)|_{q'} = (x\sigma)|_{q''}.$$

Proof of the claim. Since $l \notin \mathcal{X}$ and $l \in T(\Sigma_i, \mathcal{X})$, write $l = f(l_1, \dots, l_n)$ with $f \in \Sigma_i$. We proceed by induction on q' .

If $q' = \epsilon$: there is no strict prefix of ϵ , so q' is (vacuously) not below a variable, and $(l\sigma)|_\epsilon = l\sigma$ has root $\text{root}(l) = f \in \Sigma_i$ (substitution does not change the root of a non-variable term), matching (a) with $q' = \epsilon \in \text{Pos}(l)$.

If $q' \neq \epsilon$, write $q' = k \cdot q_1$ with $k \in \{1, \dots, n\}$ and $q_1 \in \text{Pos}(l_k\sigma)$. Since $f \notin \mathcal{X}$, the root of l is not a variable, so "below a variable" for $q' = k \cdot q_1$ reduces to "below a variable of l_k " for q_1 . By induction on the strict subterm l_k : either q_1 is not below a variable of l_k , giving $q_1 \in \text{Pos}(l_k)$ and $\text{root}((l_k\sigma)|_{q_1}) = \text{root}(l_k|_{q_1}) \in \Sigma_i$, hence $q' = k \cdot q_1 \in \text{Pos}(l)$ with the same root, matching (a); or q_1 is below a variable x of l_k at $p'_x \sqsubseteq q_1$, giving (b) for q' with $p_x = k \cdot p'_x$. This proves the claim.

Case $p = q$: this case cannot occur.

Assume $p = q$. Then $s|_p = l_1\sigma_1 = l_2\sigma_2$. Applying the preliminary claim to l_1, σ_1 at $q' = \epsilon$ (case (a), since ϵ is never below a variable) gives

$$\text{root}(l_1\sigma_1) = \text{root}(l_1) \in \Sigma_1.$$

Symmetrically, applying it to l_2, σ_2 at $q' = \epsilon$ gives $\text{root}(l_2\sigma_2) = \text{root}(l_2) \in \Sigma_2$. Since $s|_p = l_1\sigma_1 = l_2\sigma_2$, both roots must coincide, contradicting $\Sigma_1 \cap \Sigma_2 = \emptyset$. Hence $p \neq q$ whenever the two steps come from R_1 and R_2 respectively.

Case 1 (p and q are disjoint).

Assume $p \parallel q$. Since the two positions are incomparable, replacing the subterm at position p does not affect the subterm at position q . Therefore,

$$(s[r_1\sigma_1]_p)|_q = s|_q = l_2\sigma_2.$$

Hence the second rewrite step is still available after the first one:

$$s[r_1\sigma_1]_p \rightarrow_{R_2} s[r_1\sigma_1]_p[r_2\sigma_2]_q.$$

Symmetrically,

$$(s[r_2\sigma_2]_q)|_p = s|_p = l_1\sigma_1,$$

so

$$s[r_2\sigma_2]_q \rightarrow_{R_1} s[r_2\sigma_2]_q[r_1\sigma_1]_p.$$

Because p and q are disjoint, the two replacements act on disjoint subtrees of s and therefore commute:

$$s[r_1\sigma_1]_p[r_2\sigma_2]_q = s[r_2\sigma_2]_q[r_1\sigma_1]_p.$$

Let w be this common term. Then

$$t \rightarrow_{R_2} w \quad \text{and} \quad u \rightarrow_{R_1} w,$$

so the peak is joinable.

Case 2 ($p < q$).

Assume p is a strict prefix of q : the R_2 -redex occurs inside the R_1 -redex. Since $s|_p = l_1\sigma_1$, we may write $q = p \cdot q'$ with $q' \in \text{Pos}(l_1\sigma_1)$.

Apply the preliminary claim to l_1, σ_1, q' . If case (a) held, we would get $\text{root}((l_1\sigma_1)|_{q'}) \in \Sigma_1$; but $(l_1\sigma_1)|_{q'} = s|_q = l_2\sigma_2$ has root in Σ_2 (by the same argument as above, applied to l_2, σ_2 at position ϵ), a contradiction. Hence case (b) holds: there exist $x \in \mathcal{V}\text{ar}(l_1)$, p_x, q'' such that

$$l_1|_{p_x} = x, \quad q' = p_x \cdot q'', \quad (x\sigma_1)|_{q''} = l_2\sigma_2.$$

The R_2 -step only modifies the instance of x . Define σ'_1 by

$$\sigma'_1(x) = (x\sigma_1)[r_2\sigma_2]_{q''}, \quad \sigma'_1(y) = \sigma_1(y) \text{ for } y \neq x.$$

Step 1. Since $q = p \cdot p_x \cdot q''$, replacing s at position q amounts to first replacing $s|_p$ at position $p_x \cdot q''$ and then reinserting the result at p :

$$u = s[r_2\sigma_2]_q = s[(s|_p)[r_2\sigma_2]_{p_x \cdot q''}]_p = s[l_1\sigma'_1]_p.$$

Step 2. Since $l_1|_{p_x} = x$ is a variable, $l_1\sigma'_1$ still matches l_1 at the root, so $u|_p = l_1\sigma'_1$, and we may apply the rule $l_1 \rightarrow r_1$ at position p :

$$u = s[l_1\sigma'_1]_p \rightarrow_{R_1} s[r_1\sigma'_1]_p =: w.$$

Step 3. The substitutions σ_1 and σ'_1 differ only on x , and $x\sigma_1 \rightarrow_{R_2} x\sigma'_1$ is precisely the step $l_2\sigma_2 \rightarrow r_2\sigma_2$ applied at position q'' inside $x\sigma_1$. If x occurs $k \geq 0$ times in r_1 , then by stability of rewriting under substitution and under context, rewriting each of the k occurrences of $x\sigma_1$ inside $r_1\sigma_1$ one after another yields

$$r_1\sigma_1 \rightarrow_{R_2}^* r_1\sigma'_1$$

(trivially, $r_1\sigma_1 = r_1\sigma'_1$ if $k = 0$). By stability under context (the reduction occurs strictly inside $s[\cdot]_p$),

$$t = s[r_1\sigma_1]_p \rightarrow_{R_2}^* s[r_1\sigma'_1]_p = w.$$

Hence $t \rightarrow_{R_2}^* w$ and $u \rightarrow_{R_1} w$, and the peak is joinable.

Case 3 ($q < p$).

This case is symmetric to Case 2, exchanging the roles of $(R_1, l_1, r_1, \sigma_1, p)$ and $(R_2, l_2, r_2, \sigma_2, q)$. By the preliminary claim applied to l_2, σ_2 at the appropriate position, writing $p = q \cdot p'_x \cdot p''$, there exist $x' \in \mathcal{V}\text{ar}(l_2)$ with $l_2|_{p'_x} = x'$ and $(x'\sigma_2)|_{p''} = l_1\sigma_1$. Defining σ'_2 by

$$\sigma'_2(x') = (x'\sigma_2)[r_1\sigma_1]_{p''}, \quad \sigma'_2(y) = \sigma_2(y) \text{ } (y \neq x'),$$

the same argument (Steps 1–3, with R_1 and R_2 interchanged) produces a term w' such that

$$t = s[r_1\sigma_1]_p \rightarrow_{R_2} w' \quad \text{and} \quad u = s[r_2\sigma_2]_q \rightarrow_{R_1}^* w'.$$

Hence the peak is joinable.

Since the cases $p \parallel q$, $p < q$, $q < p$ are exhaustive (the case $p = q$ being excluded by the preliminary claim), every local peak $t \leftarrow s \rightarrow_{R_2} u$ is joinable. $\square$

Lemma (*extension of confluence to extra variables*). Let $R_i = (\Sigma_i, \mathcal{R}_i)$ be confluent on $\mathcal{T}(\Sigma_i, \mathcal{X})$, and let $\mathcal{X}' \supseteq \mathcal{X}$ be any (larger) set of variables. Then R_i is confluent on $\mathcal{T}(\Sigma_i, \mathcal{X}')$.

Proof. Let $C_1, C_2 \in \mathcal{T}(\Sigma_i, \mathcal{X}')$ be such that

$$C \rightarrow_{R_i}^* C_1 \quad \text{and} \quad C \rightarrow_{R_i}^* C_2$$

for some $C \in \mathcal{T}(\Sigma_i, \mathcal{X}')$. Let $\square_1, \dots, \square_n$ be the variables of $\mathcal{X}' \setminus \mathcal{X}$ occurring in C . Since every rule $l \rightarrow_R r \in \mathcal{R}_i$ satisfies $l \notin \mathcal{X}$, no rule can match a redex whose root is one of the $\square_k$; hence every rewrite step applied to C , C_1 , or C_2 occurs at a position not equal to any occurrence of a $\square_k$, and rewriting only propagates the $\square_k$'s exactly as it would propagate ordinary variables of $\mathcal{X}$ (matching and substitution treat $\mathcal{X}'$ -variables identically to $\mathcal{X}$ -variables, since no rule's left-hand side distinguishes them). Consequently, renaming $\square_1, \dots, \square_n$ to fresh variables $y_1, \dots, y_n \in \mathcal{X}$ not occurring elsewhere yields terms $C^\rho, C_1^\rho, C_2^\rho \in \mathcal{T}(\Sigma_i, \mathcal{X})$ (where ρ is the renaming) such that

$$C^\rho \rightarrow_{R_i}^* C_1^\rho \quad \text{and} \quad C^\rho \rightarrow_{R_i}^* C_2^\rho,$$

since rewriting is stable under the bijective renaming ρ . By confluence of R_i on $\mathcal{T}(\Sigma_i, \mathcal{X})$, there exists $D \in \mathcal{T}(\Sigma_i, \mathcal{X})$ such that

$$C_1^\rho \rightarrow_{R_i}^* D \quad \text{and} \quad C_2^\rho \rightarrow_{R_i}^* D.$$

Applying ρ^{-1} (again stability under renaming) gives

$$C_1 \rightarrow_{R_i}^* D^{\rho^{-1}} \quad \text{and} \quad C_2 \rightarrow_{R_i}^* D^{\rho^{-1}},$$

so C_1 and C_2 are joinable. $\square$

Lemma (*skeleton-alien commutation*). Let $t = C[s_1, \dots, s_n]$ be i -rooted with skeleton C and aliens $s_1, \dots, s_n$ at positions $p_1, \dots, p_n$. Let

$$t \xrightarrow[p]{l \rightarrow_R r} t'$$

be a single rewrite step. Exactly one of the following holds:

- (a) p occurs strictly below some p_k (i.e. $p_k < p$): then $l \rightarrow_R r \in \mathcal{R}_j$ where j is the index of s_k , the step rewrites s_k internally,

$$s_k \rightarrow_{R_j} s'_k, \quad t' = C[s_1, \dots, s'_k, \dots, s_n],$$

and this step is non-collapsing whenever t is preserved.

- (b) p does not occur below any p_k : then $l \rightarrow_R r \in \mathcal{R}_i$, and viewing C as a term over $\mathcal{T}(\Sigma_i, \mathcal{X} \cup \{\square_1, \dots, \square_n\})$ with $\square_k \mapsto s_k$,

$$C \rightarrow_{R_i} C' \quad (\text{a step of the skeleton, } \square_k \text{'s treated as variables}), \quad t' = C'[s_1, \dots, s_n]$$

unless $p = p_k$ for some k and $l \rightarrow_R r$ is collapsing, in which case $t' = s_k[?]$ (the whole alien s_k is exposed, possibly further rewritten by the substitution); this last possibility cannot occur when t is preserved.

Proof. Since $p_1, \dots, p_n$ are pairwise disjoint (Definition [Alien term], maximality), and p is a single position, exactly one of the following holds: p is strictly below some (unique) p_k ; or $p = p_k$ for some (unique) k ; or p is disjoint from every p_k , or p is a strict prefix of some p_k with $p \neq$ any ancestor collapsing into an alien boundary — but by the maximality in the alien definition, every proper prefix $q < p_k$ of an alien position satisfies $\text{root}(t|_q) \in \Sigma_i$, so if $p < p_k$ then $t|_p$ is i -rooted, hence $l \in \mathcal{R}_i$ matches $t|_p$, and this falls under case (b) with p "above" p_k (the redex properly contains the alien s_k , treated as part of the skeleton). This exhausts all cases, giving exactly (a) or (b) as stated.

Case (a). If $p_k < p$, then $t|_p = s_k|_{p'}$ for $p = p_k \cdot p'$, $p' \neq \epsilon$, so $\text{root}(t|_p) \in \Sigma_j$ (j = index of s_k), and therefore $l \rightarrow_R r \in \mathcal{R}_j$ (a rule's left-hand side and its root symbol determine which system it belongs to, since $\Sigma_1 \cap \Sigma_2 = \emptyset$). By stability of replacement under nested positions,

$$t' = t[r\sigma]_p = C[s_1, \dots, s_k[r\sigma]_{p'}, \dots, s_n] = C[s_1, \dots, s'_k, \dots, s_n],$$

where $s'_k = s_k[r\sigma]_{p'}$, i.e. $s_k \rightarrow_{R_j} s'_k$ using the same rule and position p' . If t is preserved, then by definition every step of every reduction issued from t is non-collapsing; since $s_k \rightarrow s'_k$ is (the image of) such a step, it is non-collapsing.

Case (b). If p is disjoint from, or a (non-strict) prefix of, every p_k : since $t|_p$ is matched by $l \in \mathcal{T}(\Sigma_i, \mathcal{X})$, $l \notin \mathcal{X}$, we have $\text{root}(t|_p) \in \Sigma_i$ if $p \neq p_k$ for all k , and $l \rightarrow_R r \in \mathcal{R}_i$. Writing t using its skeleton, the redex $t|_p$ lies entirely within C (with the $\square_k$'s standing in for the s_k 's at positions below p , if any p_k lies below p , unaffected by the substitution since σ only instantiates variables of $\mathcal{X}$, and the $\square_k$'s are treated as opaque constants by matching against $l \in \mathcal{T}(\Sigma_i, \mathcal{X})$, which does not mention them). Hence the step is exactly a step of C (as a term over the extended variable set), replacing C by C' , and $t' = C'[s_1, \dots, s_n]$.

If $p = p_k$ for some k : then $t|_{p_k} = s_k$ is matched by l , so $l\sigma = s_k$; if $l \rightarrow_R r$ is collapsing, $r = x \in \mathcal{X}$ and $t' = t[\sigma(x)]_{p_k}$, exposing part of s_k (or all of it) as $t'|_{p_k}$: this is a collapsing step applied to t , hence excluded when t is preserved. If $l \rightarrow_R r$ is non-collapsing, this step is absorbed into case (b) as an ordinary skeleton step at $\square_k$'s position (with $x\sigma$ instances inside r possibly re-embedding copies of s_k 's own subterms — but since $l\sigma = s_k$ entirely, and $l \in \mathcal{T}(\Sigma_i, \mathcal{X})$ while $s_k \in \mathcal{T}(\Sigma_j, \mathcal{X})$ with $\Sigma_i \cap \Sigma_j = \emptyset$, this forces $l \in \mathcal{X}$, contradicting that l is a left-hand side; hence $p = p_k$ with a non-collapsing rule cannot actually occur). $\square$

Lemma (*purity is preserved under own-system rewriting*). Let $s \in \mathcal{T}(\Sigma_j, \mathcal{X})$ be pure for R_j , and suppose $s \rightarrow_{R_j} s'$. Then $s' \in \mathcal{T}(\Sigma_j, \mathcal{X})$: s' is still pure.

Proof. Write $s \xrightarrow[\ell \rightarrow r]{\ell \rightarrow r} s'$ with $\ell \rightarrow r \in \mathcal{R}_j$, $s|_q = \ell\sigma$, $s' = s[r\sigma]_q$. By definition of a TRS, $\ell, r \in \mathcal{T}(\Sigma_j, \mathcal{X})$, so every function symbol of r belongs to Σ_j . Since s is pure, every subterm of s substituted for a variable of ℓ under σ is itself in $\mathcal{T}(\Sigma_j, \mathcal{X})$ (a subterm of a pure term is pure). Hence $r\sigma$ has all symbols in Σ_j , and replacing $s|_q$ by $r\sigma$ inside s (itself pure) yields $s' \in \mathcal{T}(\Sigma_j, \mathcal{X})$. $\square$

Lemma (*subterms of preserved terms are preserved*). Let t be preserved and $s \sqsubset t$. Then s is preserved.

Proof. Suppose towards a contradiction that some reduction sequence issued from s , $s = s_0 \rightarrow s_1 \rightarrow \dots$, contains a collapsing step, say $s_k \rightarrow s_{k+1}$ at index k . Since $s \sqsubset t$, write $t = C[s]$ for a (one-hole) context C . By stability under context, each step $s_l \rightarrow s_{l+1}$ reinjects into t as $C[s_l] \rightarrow C[s_{l+1}]$, using the same rule at the same relative position — hence the same collapsing

status. This produces a reduction sequence issued from t , namely $t = C[s_0] \rightarrow C[s_1] \rightarrow \dots$, containing the collapsing step $C[s_k] \rightarrow C[s_{k+1}]$, contradicting that t is preserved. $\square$

Definition (*strict subterm order*). Let $t \in \mathcal{T}(\Sigma, \mathcal{X})$. For $s \in \mathcal{T}(\Sigma, \mathcal{X})$, we write $s \sqsubset t$ (s is a strict subterm of t) if there exists a position $p \in \text{Pos}(t)$, $p \neq \varepsilon$, such that $s = t|_p$. We write $s \sqsubseteq t$ if $s \sqsubset t$ or $s = t$.

Lemma (*well-foundedness of the strict subterm order*). The strict subterm relation $\sqsubset$ on $\mathcal{T}(\Sigma, \mathcal{X})$ is well-founded: there is no infinite sequence $(t_k)_{k \in \mathbb{N}}$ of terms with $t_0 \sqsupset t_1 \sqsupset t_2 \sqsupset \dots$.

Proof. Let $|\cdot| : \mathcal{T}(\Sigma, \mathcal{X}) \rightarrow \mathbb{N}^*$ be defined by recursion on the structure of t (well-defined since every term is built in finitely many steps — see the remark below):

$$|t| = \begin{cases} 1 & \text{if } t \in \mathcal{X} \text{ or } t \text{ is a constant,} \\ 1 + |t_1| + \dots + |t_n| & \text{if } t = f(t_1, \dots, t_n), n \geq 1. \end{cases}$$

($|t|$ counts the total number of symbols — variables, constants, and occurrences of function symbols — occurring in t .)

Step 1: if $s \sqsubset t$, then $|s| < |t|$.

By definition of $\sqsubset$, there exists $p \in \text{Pos}(t)$, $p \neq \varepsilon$, such that $s = t|_p$. We proceed by induction on the length of p .

- If $|p| = 1$: $p = i$ for some $i \in \{1, \dots, n\}$, where $t = f(t_1, \dots, t_n)$ (with $n \geq 1$, since otherwise t would have no non-root position and p could not exist). Then $s = t|_i = t_i$, and by definition of $|\cdot|$:

$$|t| = 1 + |t_1| + \dots + |t_n| = 1 + |t_i| + \sum_{j \neq i} |t_j| \geq 1 + |t_i| > |t_i| = |s|,$$

since each $|t_j| \geq 1$ (every term has at least one symbol) and the "1" for the root symbol is added regardless. Hence $|s| < |t|$.

- If $|p| = k + 1 > 1$: write $p = i \cdot p'$ with $|p'| = k \geq 1$, $i \in \{1, \dots, n\}$. Then $s = t|_p = (t_i)|_{p'} = (t_i)|_{p'}$, so $s \sqsubset t_i$ via the position p' of length k . By the induction hypothesis (on the length of the position), $|s| < |t_i|$. By the base case above applied to $t_i \sqsubset t$ (position i , length 1), $|t_i| < |t|$. By transitivity of $<$ on $\mathbb{N}$, $|s| < |t_i| < |t|$, hence $|s| < |t|$.

This establishes Step 1 for every position length ≥ 1 , by induction.

Step 2: conclusion via well-foundedness of $<$ on $\mathbb{N}$.

Suppose, for contradiction, that an infinite sequence $t_0 \sqsupset t_1 \sqsupset t_2 \sqsupset \dots$ exists. Applying Step 1 to each $t_{k+1} \sqsubset t_k$ yields

$$|t_0| > |t_1| > |t_2| > \dots,$$

an infinite strictly decreasing sequence of positive natural numbers. But $<$ is well-founded on $\mathbb{N}$ (every non-empty subset of $\mathbb{N}$ has a least element; in particular $\{|t_k| : k \in \mathbb{N}\}$ has one, say $|t_{k_0}|$, and then $|t_{k_0+1}| < |t_{k_0}|$ contradicts minimality). Contradiction.

Hence no infinite $\sqsubset$ -decreasing sequence exists: $\sqsubset$ is well-founded. $\square$

Nota. By construction, $\mathcal{T}(\Sigma, \mathcal{X})$ is defined as the *smallest* set containing $\mathcal{X}$ and closed under $f(t_1, \dots, t_n)$ for $f \in \Sigma$ of arity n and $t_1, \dots, t_n \in \mathcal{T}(\Sigma, \mathcal{X})$ already constructed. In particular, every $t \in \mathcal{T}(\Sigma, \mathcal{X})$ is built in finitely many applications of this rule, hence has finite depth and finitely many symbols: there is no infinite term. This is what makes the recursive definition of $|\cdot|$ above well-defined (the recursion bottoms out after finitely many steps on any given t), and it is silently used throughout: a position $p \in \text{Pos}(t)$ is always a finite string, and $\text{Pos}(t)$ itself is a finite set.

Definition (*rank*). Let $t \in \mathcal{T}(\Sigma, \mathcal{X})$. We define $\text{rank}(t) \in \mathbb{N}^*$ by well-founded recursion on $\sqsubset$:

$$\text{rank}(t) = \begin{cases} 1 & \text{if } t \text{ is pure (for } R_1 \text{ or for } R_2), \\ 1 + \max\{\text{rank}(s) : s \text{ is an alien of } t\} & \text{otherwise.} \end{cases}$$

This is well defined: if t is not pure, it has at least one alien (by definition of alien / skeleton decomposition), and every alien s of t satisfies $s \sqsubset t$ strictly, so the recursion terminates by well-foundedness of $\sqsubset$.

Lemma (*strict rank decrease for aliens*). If s is an alien of t , then $\text{rank}(s) < \text{rank}(t)$.

Proof. Since s is an alien of t , t is not pure (a pure term has no alien, by Definition [Alien term]). Hence, by the second clause of Definition [rank],

$$\text{rank}(t) = 1 + \max\{\text{rank}(s') : s' \text{ alien of } t\}.$$

As s is itself one of the aliens ranging over this maximum,

$$\text{rank}(t) \geq 1 + \text{rank}(s) > \text{rank}(s).$$

□

Lemma (*reduction decomposition*). Let $t = C_0[s_1, \dots, s_n]$ be preserved, i -rooted, with skeleton C_0 and aliens $s_1, \dots, s_n$. Let $t = t_0 \rightarrow t_1 \rightarrow \dots \rightarrow t_m$ be any finite $R_1 \cup R_2$ -reduction. Then there exist, for each $k = 0, \dots, m$, an integer n_k , a skeleton $C_k \in \mathcal{T}(\Sigma_i, \mathcal{X} \cup \{\square_1, \dots, \square_{n_k}\})$, and a list $(u_1^{(k)}, \dots, u_{n_k}^{(k)})$ of aliens, with $C_0, (u_1^{(0)}, \dots) = (s_1, \dots, s_n)$ as given, such that $t_k = C_k[u_1^{(k)}, \dots, u_{n_k}^{(k)}]$ for every k , and for each $k < m$ exactly one of:

- (a) $C_{k+1} = C_k$ and there is a unique l with $u_l^{(k)} \rightarrow_{R_1 \cup R_2} u_l^{(k+1)}$, all other entries unchanged;
- (b) $C_k \rightarrow_{R_i} C_{k+1}$ (a step over $\mathcal{T}(\Sigma_i, \mathcal{X} \cup \{\square_\bullet\})$, holes treated as constants), and $(u_1^{(k+1)}, \dots)$ is $(u_1^{(k)}, \dots)$ with entries duplicated/erased according to the multiplicities of the $\square_l$'s in the step.

Consequently, discarding type-(a) steps yields $C_0 \rightarrow_{R_i}^* C_m$.

Proof. Induction on m . For $m = 0$, take $C_0, (s_1, \dots, s_n)$ as given.

Assume the decomposition built up to $t_k = C_k[u_1^{(k)}, \dots, u_{n_k}^{(k)}]$, and let $t_k \xrightarrow[p]{\ell \rightarrow r} t_{k+1}$ be the next step. Apply Lemma [skeleton-alien commutation] to t_k (viewed with skeleton C_k and aliens $u_1^{(k)}, \dots, u_{n_k}^{(k)}$): exactly one of cases (a) or (b) of that lemma holds, and the exceptional

sub-case of (b) (a collapsing step exposing $u_l^{(k)}$ at the root) cannot occur, since t_k is a term of a reduction issued from t (preserved), so no step of this reduction is collapsing. This yields case (a) or (b) above verbatim, with $C_{k+1}, (u_1^{(k+1)}, \dots)$ as described. This closes the induction.

The final claim follows since type-(a) steps leave C_k unchanged, so removing them from the sequence $C_0, C_1, \dots, C_m$ (with repeats collapsed) still gives a valid chain of type-(b) steps, i.e. a genuine reduction $C_0 \rightarrow_{R_i}^* C_m$. $\square$

Lemma (*Confluence of preserved terms*). Let $R_1 = (\Sigma_1, \mathcal{R}_1)$ and $R_2 = (\Sigma_2, \mathcal{R}_2)$ be two confluent term rewriting systems such that $\Sigma_1 \cap \Sigma_2 = \emptyset$. Let $t \in \mathcal{T}(\Sigma, \mathcal{X})$ be a preserved term. Then t is confluent for $R_1 \cup R_2$: for every t_1, t_2 such that

$$t \rightarrow_{R_1 \cup R_2}^* t_1 \quad \text{and} \quad t \rightarrow_{R_1 \cup R_2}^* t_2,$$

there exists w such that

$$t_1 \rightarrow_{R_1 \cup R_2}^* w \quad \text{and} \quad t_2 \rightarrow_{R_1 \cup R_2}^* w.$$

Proof. Let $R = R_1 \cup R_2$ and

$$T = \{ t \text{ preserved} \mid \exists t_1, t_2 : t_1 \xrightarrow{+}_R t \xrightarrow{+}_R t_2 \text{ and } \nexists w, t_1 \xrightarrow{*}_R w \xleftarrow{*}_R t_2 \}.$$

Suppose, towards a contradiction, $T \neq \emptyset$. Since rank takes values in $\mathbb{N}^*$, well-ordering of $\mathbb{N}$ gives $n_0 = \min\{\text{rank}(t) : t \in T\}$; fix $t \in T$ with $\text{rank}(t) = n_0$ (no $t' \in T$ has $\text{rank}(t') < n_0$), and fix witnesses t_1, t_2 as in the definition of T .

Case $n_0 = 1$. Then t is pure, say $t \in \mathcal{T}(\Sigma_1, \mathcal{X})$. No rule of $\mathcal{R}_2$ can match any position of t (every left-hand side of $\mathcal{R}_2$ is rooted in Σ_2 , while every symbol of t is in Σ_1), so every R -step from t is in fact an R_1 -step, and by Lemma [purity is preserved under own-system rewriting], every term reachable from t stays in $\mathcal{T}(\Sigma_1, \mathcal{X})$. Hence $t \rightarrow_R^* t_1$ and $t \rightarrow_R^* t_2$ are really R_1 -reductions, and confluence of R_1 gives w with $t_1 \rightarrow_{R_1}^* w \xleftarrow{*}_{R_1} t_2$, contradicting $t \in T$.

Case $n_0 > 1$. Then t is not pure; say t is 1-rooted with skeleton decomposition $t = C_0[s_1, \dots, s_n]$ ($n \geq 1$), aliens $s_1, \dots, s_n$ pure for Σ_2 . By Lemma [strict rank decrease for aliens], $\text{rank}(s_k) < n_0$ for every k ; by Lemma [subterms of preserved terms are preserved], each s_k is preserved. By minimality of n_0 , no preserved term of rank $< n_0$ lies in T , so each $s_k \notin T$; since s_k is preserved, this means precisely that s_k is confluent for R .

Apply Lemma [reduction decomposition] to $t \rightarrow_R^* t_1$: this gives $C_0 \rightarrow_{R_1}^* C^{(1)}$ and $t_1 = C^{(1)}[u_1^{(1)}, \dots]$, where the list $u^{(1)}$ consists of terms each reachable from some s_k by a (possibly empty) R -reduction (the type-(a) steps of the decomposition, tracing the origin k of each surviving/duplicated copy). Symmetrically, $C_0 \rightarrow_{R_1}^* C^{(2)}$ and $t_2 = C^{(2)}[u_1^{(2)}, \dots]$.

By Lemma [extension of confluence to extra variables], R_1 is confluent on $\mathcal{T}(\Sigma_1, \mathcal{X} \cup \{\square_1, \dots\})$; applied to the peak $C^{(1)} \xleftarrow{*}_{R_1} C_0 \rightarrow_{R_1}^* C^{(2)}$, this gives C^* with $C^{(1)} \rightarrow_{R_1}^* C^*$ and $C^{(2)} \rightarrow_{R_1}^* C^*$. By stability under substitution,

$$t_1 = C^{(1)}[u^{(1)}] \rightarrow_R^* C^*[u^{(1)}], \quad t_2 = C^{(2)}[u^{(2)}] \rightarrow_R^* C^*[u^{(2)}].$$

Each hole of C^* traces back (through the sequences of duplications/erasures recorded in Lemma [reduction decomposition]) to a unique origin $k \in \{1, \dots, n\}$: call $u_h^{(1)}, u_h^{(2)}$ the two fillings of hole h , both reachable from s_k by R -reductions. Since s_k is confluent, there exists

w_h with $u_h^{(1)} \rightarrow_R^* w_h \leftarrow_R^* u_h^{(2)}$. Setting $w := C^*[(w_h)_h]$ (stability under context, applied at each of the finitely many holes), we get

$$t_1 \rightarrow_R^* C^*[u^{(1)}] \rightarrow_R^* w, \quad t_2 \rightarrow_R^* C^*[u^{(2)}] \rightarrow_R^* w,$$

contradicting $t \in T$.

Both cases yield a contradiction, hence $T = \emptyset$: every preserved term is confluent for $R_1 \cup R_2$. $\square$

Lemma (*rank is non-increasing under rewriting*). If $t \rightarrow_{R_1 \cup R_2}^* t'$, then $\text{rank}(t') \leq \text{rank}(t)$.

Proof. Write $t \xrightarrow[q]{\ell \rightarrow r} t'$ with $\ell \rightarrow r \in \mathcal{R}_i$ for some $i \in \{1, 2\}$, $t|_q = \ell\sigma$, $t' = t[r\sigma]_q$. By definition of a TRS, $\ell, r \in \mathcal{T}(\Sigma_i, \mathcal{X})$: every function symbol occurring in r belongs to Σ_i . Hence $r\sigma$ is obtained from r by substituting, for each variable $x \in \text{Var}(r)$, the subterm $x\sigma$ already occurring inside t (as part of the match $t|_q = \ell\sigma$) — possibly duplicated, erased, or left in place, but never altered in content. No alien of t' can therefore exhibit an alternation pattern deeper than what was already present in some alien of t : every alien of $r\sigma$ (i.e. of $t'|_q$) is a copy of some alien already occurring inside $t|_q$, and every alien of t' outside position q is unchanged from t (Definition [term preserver]). By a straightforward structural induction on the (finite) construction of rank, it follows that $\text{rank}(t') \leq \text{rank}(t)$. $\square$

Lemma (*first-switch decomposition*). Let t be i -rooted with skeleton $t = C_0[s_1, \dots, s_n]$, and let $t \rightarrow^* t'$ be any finite $R_1 \cup R_2$ -reduction. Exactly one of the following holds:

- (N) (*no switch*) the reduction contains no collapsing step exposing the currently active root; then, by Lemma [reduction decomposition] (whose proof only requires this specific reduction to avoid root-collapses, not that t be preserved), $t' = C'[u_1, \dots, u_k]$ for some skeleton C' with $C_0 \rightarrow_{R_i}^* C'$, and each u_l a descendant of some s_j ;
- (S) (*one switch*) there is a prefix $t \rightarrow^* v = C'[u_1, \dots, u_k]$ of type (N), followed by a single collapsing step at the root exposing one filling u_{k_0} , i.e. $v \rightarrow u_{k_0}$, followed by an arbitrary further reduction $u_{k_0} \rightarrow^* t'$. In this case $\text{rank}(u_{k_0}) < \text{rank}(t)$ (Lemma [strict rank decrease for aliens]) and, by Lemma [rank is non-increasing under rewriting], $\text{rank}(t') \leq \text{rank}(u_{k_0}) < \text{rank}(t)$.

Proof. Immediate by repeated application of Lemma [skeleton-alien commutation] along the reduction $t \rightarrow^* t'$: as long as no step collapses the currently active root, every step falls under case (a) or (b) of that lemma relative to the currently active skeleton, which is precisely case (N). The first step (if any) that collapses the currently active root produces case (S); by Lemma [strict rank decrease for aliens] the rank strictly drops at that point, and by Lemma [rank is non-increasing under rewriting] it cannot climb back up afterwards, so no second "root collapse relative to the original skeleton C_0 " needs to be distinguished: from that point on we are simply reducing a term of strictly smaller rank, to which the induction hypothesis of the main theorem will apply directly. $\square$

proof of the theorem on the confluence of a union of disjoint confluent term rewriting systems.

Proof. By well-founded induction on $\text{rank}(t)$: we show, for every $n_0 \in \mathbb{N}^*$, that every term of rank n_0 is confluent for $R := R_1 \cup R_2$, assuming **(IH)** that every term of rank strictly less than n_0 is confluent for R .

Case $n_0 = 1$. Then t is pure, say $t \in \mathcal{T}(\Sigma_1, \mathcal{X})$. No rule of $\mathcal{R}_2$ can match any position of t (every left-hand side of $\mathcal{R}_2$ is rooted in Σ_2 , while every symbol of t lies in Σ_1), so every R -step issued from t is in fact an R_1 -step, and by Lemma [purity is preserved under own-system rewriting], every term reached from t remains in $\mathcal{T}(\Sigma_1, \mathcal{X})$. Confluence of t for R therefore reduces directly to confluence of R_1 , which is a hypothesis of the theorem.

Case $n_0 > 1$. Let t be i -rooted, with skeleton $t = C_0[s_1, \dots, s_n]$ ($n \geq 1$), and consider a peak $t_1 \xrightarrow{*} t \xrightarrow{*} t_2$. By Lemma [first-switch decomposition], each of the two reductions is of type (N) or (S).

Case (N,N). Both reductions stay within skeleton C_0 without ever collapsing at the active root: $t_1 = C^{(1)}[u^{(1)}]$, $t_2 = C^{(2)}[u^{(2)}]$, with $C_0 \xrightarrow{*_{R_i}} C^{(1)}$ and $C_0 \xrightarrow{*_{R_i}} C^{(2)}$. By Lemma [extension of confluence to extra variables], R_i is confluent on $\mathcal{T}(\Sigma_i, \mathcal{X} \cup \{\square_1, \dots\})$; applied to the peak $C^{(1)} \xleftarrow{*} C_0 \xrightarrow{*} C^{(2)}$, this gives C^* with $C^{(1)} \xrightarrow{*_{R_i}} C^*$, $C^{(2)} \xrightarrow{*_{R_i}} C^*$. By stability under substitution, $t_1 \xrightarrow{*} C^*[u^{(1)}]$, $t_2 \xrightarrow{*} C^*[u^{(2)}]$. Each hole h of C^* traces back to a unique originating alien s_k ($k \in \{1, \dots, n\}$); by Lemma [strict rank decrease for aliens], $\text{rank}(s_k) < n_0$, so by **(IH)** s_k is confluent for R : we join the two fillings of hole h (both reducts of s_k) into a term w_h . Setting $w := C^*[(w_h)_h]$ (stability under context, applied at each of the finitely many holes), we obtain $t_1 \xrightarrow{*_R} w$ and $t_2 \xrightarrow{*_R} w$: the peak is joinable.

Case (N,S) (case (S,N) is symmetric). Write $t_1 = C^{(1)}[u^{(1)}]$ (type N), and for branch 2: $t \xrightarrow{*} v = C^{(2)}[u^{(2)}]$ (type N up to this point), followed by a single collapsing step at the root exposing one filling $u_{k_0}^{(2)}$, i.e. $v \rightarrow u_{k_0}^{(2)}$, followed by $u_{k_0}^{(2)} \xrightarrow{*} t_2$, with $\text{rank}(u_{k_0}^{(2)}) < n_0$ (Lemma [first-switch decomposition]).

We first join t_1 and v exactly as in case (N,N) above (both are type-(N) reductions issued from t within skeleton C_0): this gives $y = C^*[y_1, \dots, y_n]$ with $t_1 \xrightarrow{*_R} y$, $v \xrightarrow{*_R} y$, where each y_l results from joining (via **IH**) the two fates of alien s_l along each branch. In particular, for $l = k_0$: the fate of s_{k_0} on branch 2 is precisely $u_{k_0}^{(2)}$ (the skeleton reduction $C^{(2)} \xrightarrow{*_{R_i}} C^*$ never modifies the content of the holes, the $\square_l$ being treated as opaque constants by R_i), so y_{k_0} is the common reduct obtained by joining $u_{k_0}^{(2)}$ with its counterpart on branch 1.

Since $\text{rank}(u_{k_0}^{(2)}) < n_0$, **(IH)** gives confluence of $u_{k_0}^{(2)}$ for R : applied to the peak $t_2 \xleftarrow{*} u_{k_0}^{(2)} \xrightarrow{*} y_{k_0}$ (the first arrow being the given $u_{k_0}^{(2)} \xrightarrow{*} t_2$, the second arising from the construction of y_{k_0} above), it yields z with $t_2 \xrightarrow{*_R} z$ and $y_{k_0} \xrightarrow{*_R} z$.

Setting $w := C^*[y_1, \dots, z, \dots, y_n]$ (replacing hole k_0 by z ; stability under context), we obtain $t_1 \xrightarrow{*_R} y \xrightarrow{*_R} w$ and, by construction of z from t_2 , $t_2 \xrightarrow{*_R} w$: the peak is joinable.

Case (S,S). Same construction, treating both collapses symmetrically: first join the two type-(N) prefixes as in case (N,N), identify for each branch the alien exposed by its own collapse, join them separately with their counterpart in the common skeleton C^* via **IH** (rank strictly less than n_0 on both sides), then recombine within C^* as above.

In every case, we obtain w such that $t_1 \xrightarrow{*_R} w$ and $t_2 \xrightarrow{*_R} w$: every peak issued from t is joinable, hence t is confluent for R . By well-founded induction on rank, every term of $\mathcal{T}(\Sigma, \mathcal{X})$ is confluent for R , i.e. $R_1 \cup R_2$ is confluent. $\square$

Nota (modularity). A property P of term rewriting systems is said to be *modular* if, whenever $R_1 = (\Sigma_1, \mathcal{R}_1)$ and $R_2 = (\Sigma_2, \mathcal{R}_2)$ both have property P and $\Sigma_1 \cap \Sigma_2 = \emptyset$, the

union $R_1 \cup R_2$ also has property P . The theorem we have just proved states precisely that *confluence is a modular property* of TRS with disjoint signatures.

This modularity is far from automatic, and it is worth contrasting it with properties for which it fails. The most instructive counterexample is *termination*, which is *not* modular: there exist two disjoint, individually terminating TRS whose union admits an infinite reduction (Toyama, 1987). A standard example takes

$$\mathcal{R}_1 = \{ f(x, y, x) \rightarrow f(x, y, y) \}, \quad \mathcal{R}_2 = \{ g(x, y) \rightarrow x, \quad g(x, y) \rightarrow y \},$$

with $\Sigma_1 = \{f\} \cup \{a, b\}$, $\Sigma_2 = \{g\}$ (constants a, b shared as needed by adding them to Σ_1 , or introduced fresh). Each of R_1, R_2 terminates on its own: R_1 has a single rule whose right-hand side is "smaller" in an appropriate sense, and R_2 strictly decreases the size of any term. Yet the union admits the infinite reduction

$$f(g(a, b), g(a, b), g(a, b)) \rightarrow f(g(a, b), g(a, b), a) \rightarrow f(a, g(a, b), a) \rightarrow f(a, g(a, b), b) \rightarrow \dots$$

obtained by repeatedly using the two collapsing rules of $\mathcal{R}_2$ to rewrite the first and third occurrences of $g(a, b)$ inconsistently, feeding the result back into $\mathcal{R}_1$'s rule and re-duplicating the alien $g(a, b)$ ad infinitum. The crucial mechanism at play is exactly the one isolated by our notion of *collapsing rule*, both rules of $\mathcal{R}_2$ collapse a term to one of its own arguments, which is precisely what allows an alien to be repeatedly re-exposed and duplicated across the two signatures without ever terminating.

This is exactly why our proof of confluence required tracking collapsing steps so carefully, and why we could not simply invoke Newman's Lemma: $R_1 \cup R_2$ need not terminate even when R_1 and R_2 both do, so local confluence together with termination (Newman's classical route to confluence) is unavailable in general. Our argument instead proceeds by well-founded induction on rank a measure entirely internal to the static structure of a term, insensitive to whether the rewriting relation terminates which is what allows the proof to go through regardless of the (non-)termination of $R_1 \cup R_2$.

Proposition (*Terminating modularity*). Termination is not a modular property.

Proof. We exhibit two disjoint terminating TRS whose union does not terminate (Toyama, 1987). Let $\Sigma_1 = \{F, 0, 1\}$ (F ternary, $0, 1$ constants) and $\Sigma_2 = \{G\}$ (G binary), with $\Sigma_1 \cap \Sigma_2 = \emptyset$, and set

$$\begin{aligned} R_1 &= (\Sigma_1, \mathcal{R}_1), \quad \mathcal{R}_1 = \{ F(0, 1, x) \rightarrow F(x, x, x) \}, \\ R_2 &= (\Sigma_2, \mathcal{R}_2), \quad \mathcal{R}_2 = \{ G(x, y) \rightarrow x, \quad G(x, y) \rightarrow y \}. \end{aligned}$$

R_2 *terminates*. Let $|\cdot|$ be the size measure of Definition [well-foundedness of the strict subterm order]. For any step $s \xrightarrow[q]{} s'$ using a rule of $\mathcal{R}_2$, the redex $G(x, y)\sigma$ is replaced by $x\sigma$ or $y\sigma$, a strict subterm of the redex (the symbol G is erased and only one of the two arguments is kept). Hence $|s'| < |s|$ strictly. Since $|\cdot|$ takes values in $\mathbb{N}_{>0}$ and $<$ is well-founded on $\mathbb{N}$, no infinite R_2 -reduction can exist.

R_1 *terminates*. A naive size argument fails here, since the rule $F(0, 1, x) \rightarrow F(x, x, x)$ triples the occurrences of x and can strictly increase the size of a term. The correct argument (Toyama, 1987; see also Drosten, 1989, for an alternative proof via a polynomial interpretation) rests on the fact that neither 0 nor 1 can ever be freshly created by $\mathcal{R}_1$: the right-hand side $F(x, x, x)$ only duplicates the subterm already substituted for x , so every occurrence

of the pattern $F(0, 1, \cdot)$ appearing after a step was already present, in duplicated form, before that step and hence, tracing back, already present in the original term. A minimal-counterexample argument on the depth of the first redex used along a hypothetical infinite reduction then yields a contradiction. We admit this result here without full detail, as termination of R_1 alone is not the subject of the present article; see the cited references for the complete proof.

The union $R_1 \cup R_2$ does not terminate. Let

$$t_0 := F(G(0, 1), G(0, 1), G(0, 1)).$$

Using $G(x, y) \rightarrow x$ on the first occurrence of $G(0, 1)$:

$$t_0 \xrightarrow{1} F(0, G(0, 1), G(0, 1)) =: t_1.$$

Using $G(x, y) \rightarrow y$ on the (now second) occurrence of $G(0, 1)$:

$$t_1 \xrightarrow{2} F(0, 1, G(0, 1)) =: t_2.$$

The term $t_2 = F(0, 1, G(0, 1))$ matches the left-hand side of the rule of $\mathcal{R}_1$ with $\sigma(x) = G(0, 1)$:

$$t_2 \xrightarrow[\varepsilon]{F(0, 1, x) \rightarrow F(x, x, x)} F(G(0, 1), G(0, 1), G(0, 1)) = t_0.$$

Hence $t_0 \rightarrow t_1 \rightarrow t_2 \rightarrow t_0$: this is a cyclic reduction, which repeats indefinitely,

$$t_0 \rightarrow t_1 \rightarrow t_2 \rightarrow t_0 \rightarrow t_1 \rightarrow t_2 \rightarrow \dots,$$

giving an infinite $R_1 \cup R_2$ -reduction issued from t_0 .

Both R_1 and R_2 terminate individually and $\Sigma_1 \cap \Sigma_2 = \emptyset$, yet $R_1 \cup R_2$ admits an infinite reduction and therefore does not terminate. This shows that termination is not a modular property. $\square$

Theorem (*Union convergent*). The union of two convergent rewrite systems sharing no function symbols (or constants), and in which no variable appears more than once on any left-hand side, is also convergent.

Proof. We realize the disjoint union of two confluent TRSs, by the previous result this union is itself confluent, now let's see why the additional condition of this theorem compared to the other gives the ending. $\square$

4.1 Orthogonal systems

Definition (*Orthogonal system*). A term rewriting system R is called *orthogonal* if it satisfies the following two properties:

1. (*non-overlapping*) For any two rules $l_1 \rightarrow r_1$ and $l_2 \rightarrow r_2$ in R , there is no overlap, i.e., there do not exist a non-variable position $p \in \text{Pos}(l_2)$ and a substitution σ such that

$$l_2|_p = l_1\sigma.$$

2. (*left-linearity*) Every left-hand side l of a rule in R is linear, i.e., no variable occurs

more than once in l .

Lemma (*unique redex identification under orthogonality*). Let R be orthogonal and $p \in \text{Pos}(t)$ such that $t|_p$ is a redex (definition of Redex). Then the rule $l \rightarrow r \in R$ and the substitution σ with $t|_p = l\sigma$ are uniquely determined by t and p .

Proof. Uniqueness of the rule. Suppose $t|_p = l_1\sigma_1 = l_2\sigma_2$ for rules $l_1 \rightarrow r_1, l_2 \rightarrow r_2 \in R$. Then $l_2|_{\varepsilon}\sigma_2 = l_1\sigma_1$: taking $p' = \varepsilon \in \text{Pos}(l_2)$ (non-variable, since $l_2 \notin \mathcal{X}$), this is an overlap of $l_1 \rightarrow r_1$ into $l_2 \rightarrow r_2$ at the root, forbidden by non-overlapping unless $l_1 \rightarrow r_1 = l_2 \rightarrow r_2$ (non-overlapping only excludes overlaps between distinct rules, or a rule overlapping itself at a non-root position; the trivial root self-overlap of a rule with itself is not excluded). Hence the rule is unique.

Uniqueness of σ . Writing $l := l_1 = l_2$, left-linearity of R ensures each $x \in \text{Var}(l)$ occurs at a single position $p_x \in \text{Pos}(l)$, forcing $\sigma(x) = t|_{p \cdot p_x}$: no two occurrences of x can impose conflicting values. $\square$

Lemma ($\Rightarrow_R^*$ equals $\rightarrow_R^*$). $\Rightarrow_R^* = \rightarrow_R^*$.

Proof. By the note following definition of Parallel rewriting, $\rightarrow_R \subseteq \Rightarrow_R \subseteq \rightarrow_R^*$. Taking reflexive-transitive closures gives $\rightarrow_R^* \subseteq \Rightarrow_R^* \subseteq (\rightarrow_R^*)^* = \Rightarrow_R^*$, hence $\Rightarrow_R^* = \rightarrow_R^*$. $\square$

Definition (*complete development*). For R orthogonal, define $^\circ : \mathcal{T}(\Sigma, \mathcal{X}) \rightarrow \mathcal{T}(\Sigma, \mathcal{X})$ by well-founded recursion on $\sqsubset$:

$$t^\circ = \begin{cases} t & \text{if } t \in \mathcal{X}, \\ r\sigma^\circ & \text{if } t|_\varepsilon \text{ is a redex, } t = l\sigma l \rightarrow r, \sigma^\circ(x) := (x\sigma)^\circ \text{ for } x \in \text{Var}(l), \\ f(t_1^\circ, \dots, t_k^\circ) & \text{if } t = f(t_1, \dots, t_k) \text{ is not itself a redex.} \end{cases}$$

Well defined: each recursive call is on a strict subterm ($x\sigma \sqsubset t$, or $t_i \sqsubset t$); the cases are mutually exclusive and exhaustive; σ° is well-defined as a substitution by left-linearity of l .

Lemma (*strip lemma*). Let R be orthogonal. If $t \Rightarrow_R t'$, then $t' \Rightarrow_R t^\circ$.

Proof. By induction on $\sqsubset$ applied to t . Let $t \Rightarrow_R t'$ via $P = \{p_1, \dots, p_n\}$, rules $l_i \rightarrow r_i$, substitutions σ_i (definition of Parallel rewriting).

Case $\varepsilon \notin P$, $t|_\varepsilon$ not a redex. Then $t = f(t_1, \dots, t_k)$, $t^\circ = f(t_1^\circ, \dots, t_k^\circ)$. Partitioning P by argument index and applying the induction hypothesis to each $t_j \sqsubset t$ gives $t' \Rightarrow_R t^\circ$ argument-wise, hence globally.

Case $\varepsilon \notin P$, $t|_\varepsilon$ is a redex. Then $t = l\sigma$ for the unique root-matching rule $l \rightarrow r$, $t^\circ = r\sigma^\circ$. Since $\varepsilon \notin P$, non-overlapping forces every $p_i \in P$ to lie strictly inside some instance $x\sigma$, $x \in \text{Var}(l)$ (left-linearity making these instances pairwise disjoint). This gives, for each such x , a parallel step $x\sigma \Rightarrow_R x\sigma'$; the induction hypothesis (applied to $x\sigma \sqsubset t$) gives $x\sigma' \Rightarrow_R (x\sigma)^\circ = \sigma^\circ(x)$. Transporting these steps through the fixed skeleton l (stability under context/substitution) gives $t' \Rightarrow_R r\sigma^\circ = t^\circ$.

Case $\varepsilon \in P$. Since P is an antichain and $\varepsilon \leq$ every position, $P = \{\varepsilon\}$, so $t' = r\sigma$ (using the same unique rule as in definition of complete development). For each $x \in \text{Var}(l)$, $x\sigma \sqsubset t$, so

applying the induction hypothesis to the trivial step $x\sigma \Rightarrow_R x\sigma$ gives $x\sigma \Rightarrow_R (x\sigma)^\circ = \sigma^\circ(x)$; combining these at the (disjoint, by left-linearity of r 's variable occurrences inherited from l) positions of each variable in $r\sigma$ gives $r\sigma \Rightarrow_R r\sigma^\circ = t^\circ$.

In every case, $t' \Rightarrow_R t^\circ$. $\square$

Lemma (*diamond property for $\Rightarrow_R$*). Let R be orthogonal. If $t \Rightarrow_R t_1$ and $t \Rightarrow_R t_2$, there exists w with $t_1 \Rightarrow_R w$ and $t_2 \Rightarrow_R w$.

Proof. By the strip lemma applied to each step: $t_1 \Rightarrow_R t^\circ$ and $t_2 \Rightarrow_R t^\circ$. Take $w := t^\circ$. $\square$

Theorem (*Orthogonal confluence (Rosen)*). Every orthogonal system is confluent.

Proof. By the lemma stating $\Rightarrow_R^* = \rightarrow_R^*$, it suffices to show $\Rightarrow_R$ is confluent. Let

$$t = u_0 \Rightarrow_R \cdots \Rightarrow_R u_m = t_1, \quad t = v_0 \Rightarrow_R \cdots \Rightarrow_R v_k = t_2.$$

We show, by induction on $m + k$, that u_m, v_k are joinable by $\Rightarrow_R^*$.

If $m = 0$ or $k = 0$, one sequence is trivial and the other provides the join directly.

If $m, k \geq 1$: by the diamond property applied to $u_0 \Rightarrow_R u_1$ and $v_0 \Rightarrow_R v_1$ (both issued from $t = u_0 = v_0$), there is z with $u_1 \Rightarrow_R z$, $v_1 \Rightarrow_R z$. Applying the induction hypothesis to the peak $t_1 \xleftarrow{*} u_1 \Rightarrow_R z$ (total length $< m + k$) gives w_1 with $t_1 \Rightarrow_R^* w_1$, $z \Rightarrow_R^* w_1$. Applying it again to the peak $w_1 \xleftarrow{*} z \Rightarrow_R^* t_2$ (via $v_1 \Rightarrow_R z \Rightarrow_R^* t_2$, still of total length $< m + k$) gives w with $w_1 \Rightarrow_R^* w$, $t_2 \Rightarrow_R^* w$. Hence $t_1 \Rightarrow_R^* w_1 \Rightarrow_R^* w$ and $t_2 \Rightarrow_R^* w$, closing the induction.

So $\Rightarrow_R$, hence $\rightarrow_R$, is confluent: every orthogonal TRS is confluent. $\square$

5 Completion

Not all TRSs are naturally confluent. Sometimes it is necessary to transform a TRS in order to obtain a confluent TRS (by adding rules, deleting rules, etc.), when this is possible, because sometimes it is not. In this part, we are going to look at a method called “completion”, or “Knuth–Bendix completion”.

Definition (*Equivalent TRS*). Let R and S be two TRSs. We say that R and S are *equivalent* if and only if $\rightarrow_R^* = \rightarrow_S^*$, that is:

$$\forall t, u \in \mathcal{T}(\Sigma, \mathcal{X}), \quad t \rightarrow_R^* u \iff t \rightarrow_S^* u.$$

Definition (*Reduction order*). A *reduction order* on $\mathcal{T}(\Sigma, \mathcal{X})$ is a strict order $\succ$ that is

- (i) well-founded,
- (ii) stable under substitution ($s \succ t \Rightarrow s\sigma \succ t\sigma$),
- (iii) stable under context ($s \succ t \Rightarrow C[s] \succ C[t]$ for every context C).

A rule $l \rightarrow r$ is *compatible* with $\succ$ if $l \succ r$;

A TRS R is compatible with $\succ$ if every rule is.

Lemma (*compatible TRS terminate*). If R is compatible with a reduction order $\succ$, then R terminates.

Proof. An infinite reduction $t_0 \rightarrow t_1 \rightarrow \dots$ would give, via stability under substitution and context at each redex, $t_0 \succ t_1 \succ \dots$, an infinite $\succ$ -decreasing sequence, contradicting well-foundedness of $\succ$. $\square$

6 Undecidability via Turing Machines

This section is independent from the rest of the paper: it illustrates, via an encoding of Turing machines as term rewriting systems, that termination of an arbitrary TRS is undecidable. It may be skipped by a reader only interested in the confluence-theoretic development of the following sections.

Definition (*Turing machine*). A *deterministic Turing machine* is a tuple

$$M = (Q, \Sigma, \Gamma, \delta, q_0, q_{\text{acc}}, q_{\text{rej}}),$$

where:

- Q is a finite set of states;
- Σ is the finite input alphabet, with $\sqcup \notin \Sigma$, where $\sqcup$ denotes the blank symbol;
- Γ is the tape alphabet such that

$$\Sigma \subseteq \Gamma \quad \text{and} \quad \sqcup \in \Gamma;$$

- $q_0 \in Q$ is the initial state;
- $q_{\text{acc}} \in Q$ is the accepting state;
- $q_{\text{rej}} \in Q$ is the rejecting state, with

$$q_{\text{acc}} \neq q_{\text{rej}};$$

- δ is the transition function

$$\delta : (Q \setminus \{q_{\text{acc}}, q_{\text{rej}}\}) \times \Gamma \longrightarrow Q \times \Gamma \times \{L, R\},$$

where L and R denote a move of the tape head one cell to the left and to the right, respectively.

Remark. We restrict attention to *deterministic* Turing machines throughout this work, since this is the setting needed for the reduction to term rewriting (definition of Turing machine) and for the undecidability results that follow. The non-deterministic case is not treated here.

Modelling of the band of a TM We assume three distinct symbols $0, 1, \sqcup$ denoting respectively a value of 0, 1, or a blank value. A fourth symbol $\sqcup_!$ denotes an infinite sequence of blank symbols. By definition and from any position, reading a tape from a starting position to the left (or to right) encounters a finite sequence of 0, 1, or blank positions, before arriving at an infinite number of blanks. Denoting $c(x, l')$ a sequence starting with the symbol x and continuing with a sequence l , the tape can be modeled with a term $\text{tape}(l, c, r)$ with l, r denoting the sequences to be read on the left and on the right, and $c \in \{0, 1, \sqcup\}$ is the symbol in the current position. Both l and r are either $\sqcup_!$ or of the form $c(x, y)$, with $x \in \{0, 1, \sqcup\}$ and y a term denoting the subsequence after the first element. This representation is in general not unique, because an infinite sequence of blanks can be denoted

with any $c(\square, c(\dots, c(\square, \square!) \dots))$. One can obtain a unique representation by "removing" the unnecessary blank symbols before the infinite sequence of blanks. This operation is performed with the *band normalisation* rewriting rule:

$$c(\square, \square!) \rightarrow \square!$$

It is clear that this rewriting system is terminating and confluent. The moves on the tape are modeled with the rewriting rules:

$$\begin{aligned} \text{left}(\text{tape}(c(x, y), x', y')) &\rightarrow \text{tape}(y, x, c(x', y')) \\ \text{right}(\text{tape}(x, y, c(x', y'))) &\rightarrow \text{tape}(c(x, y), x', y') \end{aligned}$$

We note that there are two critical pairs with each of these rules and the band normalisation rule. These pairs are:

$$\begin{aligned} (\text{left}(\text{tape}(\square!, x', y')), \text{tape}(\square!, \square, c(x', y'))) \\ (\text{right}(\text{tape}(x, y, \square!)), \text{tape}(c(x, y), \square, \square!)) \end{aligned}$$

Using the Knuth-Bendix algorithm, we *orient* these pairs into two additional rewriting rules:

$$\begin{aligned} \text{left}(\text{tape}(\square!, x', y')) &\rightarrow \text{tape}(\square!, \square, c(x', y')) \\ \text{right}(\text{tape}(x, y, \square!)) &\rightarrow \text{tape}(c(x, y), \square, \square!) \end{aligned}$$

to obtain a convergent rewriting system. We denote by R_{tape} this convergent system (band normalisation, together with the four move rules above), acting on $\text{tape}(\cdot, \cdot, \cdot)$ -terms and their images under $\text{left}(\cdot), \text{right}(\cdot)$.

Proposition (*termination of the band normalisation rule*). The rewriting system $R_{\square} = \{ c(\square, \square!) \rightarrow \square! \}$ is terminating.

Proof. Each rewrite step replaces a subterm $c(\square, \square!)$, made of 3 function symbols, by the term $\square!$, made of 1 function symbol. So each step strictly decreases the total number of function symbols in the term.

Since this number is a natural number, it cannot decrease infinitely many times. Hence there is no infinite reduction sequence, and $R_{\square}$ is terminating. $\square$

Proposition (*confluence of the band normalisation rule*). The rewriting system $R_{\square} = \{ c(\square, \square!) \rightarrow \square! \}$ is confluent.

Proof. Since $R_{\square}$ is terminating, it suffices, by Newman's lemma, to show local confluence.

Let $t \rightarrow_{R_{\square}} b$ and $t \rightarrow_{R_{\square}} c$, using the rule at positions p and q respectively.

If $p = q$, then $b = c$, so the peak is trivially joinable.

If $p \neq q$, the two positions cannot be nested: the term $c(\square, \square!)$ has no strict subterm equal to $c(\square, \square!)$ itself (its only subterms are the constants $\square$ and $\square!$). So p and q are disjoint, and the two rewrite steps do not interfere: we can apply both, in either order, and reach the same term d .

Hence every peak is joinable, $R_{\square}$ is locally confluent, and by Newman's lemma, it is confluent. $\square$

Definition (*input term*). Let $\Sigma \subseteq \Gamma$ be the input alphabet of M . We define the set I_Σ of *input terms* over Σ by induction:

- $\square_! \in I_\Sigma$;
- if $w \in I_\Sigma$ and $a \in \Sigma$, then $c(a, w) \in I_\Sigma$.

Every input term $w \in I_\Sigma$ is thus, by construction, of one of the two following forms: either $w = \square_!$, or there exist $n \geq 1$, symbols $a_1, \dots, a_n \in \Sigma$, such that

$$w = c(a_1, c(a_2, \dots, c(a_n, \square_!) \dots)).$$

Definition (*configuration term*). Let $M = (Q, \Sigma, \Gamma, \delta, q_0, q_{\text{acc}}, q_{\text{rej}})$ be a deterministic Turing machine. We reuse the term $\text{tape}(l, c, r)$ introduced above, representing the content of the tape around the head, where l, r are each either $\square_!$ or of the form $c(x, y)$ with $x \in \Gamma$, and $c \in \Gamma$ is the symbol under the head.

A *configuration* of M is a term

$$\text{conf}(q, \text{tape}(l, c, r)),$$

where $q \in Q$.

Given an input term $w \in I_\Sigma$ with $w = c(a_1, c(a_2, \dots, c(a_n, \square_!) \dots))$ for some $n \geq 1$ and $a_1, \dots, a_n \in \Sigma$, the *initial configuration* of M on input w is the term

$$t_{M,w} = \text{conf}(q_0, \text{tape}(\square_!, a_1, c(a_2, \dots, c(a_n, \square_!) \dots))).$$

Definition (*transition rules*). Let $M = (Q, \Sigma, \Gamma, \delta, q_0, q_{\text{acc}}, q_{\text{rej}})$ be a deterministic Turing machine. For every state $q \in Q \setminus \{q_{\text{acc}}, q_{\text{rej}}\}$ and every symbol $a \in \Gamma$, write

$$\delta(q, a) = (q', a', d), \quad d \in \{L, R\}.$$

We define the rewriting system R_M over configurations, containing, for every such pair (q, a) , exactly one rule:

- if $d = R$:

$$\text{conf}(q, \text{tape}(l, a, r)) \longrightarrow \text{right}(\text{conf}(q', \text{tape}(l, a', r)));$$

- if $d = L$:

$$\text{conf}(q, \text{tape}(l, a, r)) \longrightarrow \text{left}(\text{conf}(q', \text{tape}(l, a', r))).$$

We further add, for every state q and every configuration $\text{conf}(q, \cdot)$, the rule

$$\text{right}(\text{conf}(q, T)) \longrightarrow \text{conf}(q, \text{right}(T)), \quad \text{left}(\text{conf}(q, T)) \longrightarrow \text{conf}(q, \text{left}(T)),$$

so that the head-move rules $\text{left}(\cdot), \text{right}(\cdot)$ introduced earlier for $\text{tape}(\cdot, \cdot, \cdot)$ act on the tape component T of the configuration, and pull the resulting new configuration back out of the left / right wrapper.

R_M is the union of $R_\square$ (band normalisation), the tape-move rules $\{\text{left}, \text{right}\}$ (applied to tape terms), the two rules above (applied to conf terms wrapped in $\text{left} / \text{right}$), and the transition rules just defined.

Definition (*halting configuration*). Let M be a deterministic Turing machine and R_M the associated rewriting system. A configuration $\text{conf}(q, \text{tape}(l, c, r))$ is called a *halting configuration* if $q \in \{q_{\text{acc}}, q_{\text{rej}}\}$.

Lemma (*state and current symbol are unaffected by $R_{\square}$*). Let $C = \text{conf}(q, \text{tape}(l, c, r))$ and let $C \rightarrow_{R_{\square}} C_1$ be a single band normalisation step. Then $C_1 = \text{conf}(q, \text{tape}(l', c, r'))$ for some l', r' : the state q and the current symbol c are unchanged.

Proof. The rule of $R_{\square}$ has left-hand side $c(\square, \square_l)$, rooted by the symbol c (the list constructor). Since $\text{root}(C) = \text{conf} \neq c$, the redex position p satisfies $p \neq \epsilon$. Since C has exactly two immediate subterms, q (at position 1) and $\text{tape}(l, c, r)$ (at position 2), and q is an element of the finite set Q (hence not decomposable, containing no subterm of the form $c(\square, \square_l)$), the redex position p must lie inside $\text{tape}(l, c, r)$, i.e. $p = 2 \cdot p'$ for some p' .

Similarly, within $\text{tape}(l, c, r)$, the current symbol c occupies position 2 (the second immediate argument of tape) and is a single element of Γ : it contains no subterm of the form $c(\square, \square_l)$ either. Hence p' lies inside l (position $2 \cdot 1$) or inside r (position $2 \cdot 3$), never at the position of c itself. Replacing at p therefore leaves both q and c unchanged, and only modifies l or r . $\square$

Definition (*computation of M*). Let M be a deterministic Turing machine and $w \in I_{\Sigma}$. We define, by induction on $k \geq 0$, a (possibly infinite) sequence of configurations $C_0, C_1, C_2, \dots$ by

$$C_0 = t_{M,w},$$

and, as long as $C_k = \text{conf}(q, \text{tape}(l, c, r))$ is not a halting configuration, writing $\delta(q, c) = (q', c', d)$:

$$C_{k+1} = \text{conf}(q', T^*),$$

where T is the tape term obtained from $\text{tape}(l, c', r)$ by one application of the move rule for direction d (as in definition transition rules, stripped of its conf -wrapper), and T^* is the (unique, by confluence and termination of R_{tape}) irreducible form of T under R_{tape} . We say M *halts* on w if every sequence of configurations **modifié, car si la machine n'est pas déterministe, il peut y avoir plusieurs séquences** Ok. La MT est supposée déterministe par définition donc il n'y a qu'une seule séquence possible sauf si je me trompe quelque part. is finite, i.e. there exists n such that C_n is a halting configuration.

Lemma (*transition subsequence*). Let

$$t_{M,w} = D_0 \rightarrow_{R_M} D_1 \rightarrow_{R_M} \dots \rightarrow_{R_M} D_N$$

be a finite reduction sequence in R_M . Let $i_1 < i_2 < \dots < i_m$ be the indices of the steps among $D_0 \rightarrow D_1, \dots, D_{N-1} \rightarrow D_N$ that are transition steps (as opposed to steps of $R_{\square}$ or of the wrapper/move rules internal to R_{tape}). Then, for every $j \in \{1, \dots, m\}$, the configuration reached immediately after the j -th transition step, namely D_{i_j+1} , coincides with C_j from Definition [computation of M] up to normalisation of its tape by R_{tape} ; more precisely,

$$D_{i_j+1} \rightarrow_{R_{\text{tape}}}^* C_j,$$

and the state and current symbol of D_{i_j+1} coincide with those of C_j .

Proof. By induction on j .

Base case $j = 0$ (before any transition). By lemma state and current symbol are unaffected by $R_\square$, every step among $D_0 \rightarrow \dots \rightarrow D_{i_1}$ (all being $R_\square$ -steps, by definition of i_1 as the first transition index) leaves the state and current symbol of D_0 unchanged. Hence D_{i_1} has the same state q_0 and current symbol c_0 as $D_0 = t_{M,w} = C_0$, and (by confluence and termination of $R_\square$, giving a unique normal form) D_{i_1} 's tape reduces, under $R_\square$, to the same normal form as C_0 's tape.

Inductive step. Assume the claim holds for $j-1$: $D_{i_{j-1}+1}$ has the same state q_{j-1} and current symbol c_{j-1} as C_{j-1} , and its tape is an $R_\square$ -reduct of C_{j-1} 's tape. The steps $D_{i_{j-1}+1} \rightarrow \dots \rightarrow D_{i_j}$ are all $R_\square$ -steps (by definition of i_j as the next transition index), so by Lemma [state and current symbol are unaffected by $R_\square$], D_{i_j} still has state q_{j-1} and current symbol c_{j-1} , with a tape that is a further $R_\square$ -reduct.

The transition rule matching D_{i_j} at the root is therefore determined by $\delta(q_{j-1}, c_{j-1})$ the same pair as for C_{j-1} and produces $D_{i_j+1} = \text{conf}(q_j, T)$ for the corresponding new state q_j and some tape term T , exactly as in definition computation of M (the move rule applied only depends on the outer shape of the relevant list, l or r , and by the convergence of R_{tape} established via Knuth-Bendix completion, this outer shape and the resulting current symbol after the move do not depend on how much of D_{i_j} 's tape has already been normalised by $R_\square$: any two tapes related by an $R_\square$ -reduction produce, after the same move rule, results that are again related by an $R_\square$ -reduction, since R_{tape} is confluent and stable under context).

Hence D_{i_j+1} and C_j have the same state q_j and current symbol c_j , and their tapes are related by R_{tape} -reduction (to the common normal form $T^* = C_j$'s tape, by uniqueness of normal forms). This proves the inductive step. $\square$

Proposition (*Turing machine by a TRS*). Let M be a deterministic Turing machine, R_M its associated rewriting system, and $w \in I_\Sigma$ an input term. Then M halts on w if and only if $t_{M,w}$ reduces, in R_M , to an irreducible term.

Proof. ($\Rightarrow$) Assume M halts on w , with halting configuration C_n reached after n steps (definition computation of M). By construction of R_M , each step $C_k \rightarrow C_{k+1}$ corresponds to exactly one transition step of R_M (rewriting the current symbol and invoking the move rule), possibly followed by finitely many $R_\square$ -steps normalising the resulting tape (to reach T^*). Concatenating these n blocks of steps gives a finite reduction $t_{M,w} \rightarrow_{R_M}^* C_n$. Since C_n is a halting configuration, its state is q_{acc} or q_{rej} , so no transition rule matches it at the root (definition transition rules only defines rules for $q \notin \{q_{\text{acc}}, q_{\text{rej}}\}$); and its tape, being the normal form T^* , contains no $R_\square$ -redex. Hence C_n is irreducible, and $t_{M,w}$ reduces to an irreducible term.

($\Leftarrow$) Assume

$$t_{M,w} = D_0 \rightarrow_{R_M} D_1 \rightarrow_{R_M} \dots \rightarrow_{R_M} D_N$$

with D_N irreducible. Let $i_1 < \dots < i_m$ be the indices of the transition steps in this sequence (lemma transition subsequence), and let q_m, c_m be the state and current symbol of D_{i_m+1} (or of D_0 if $m = 0$, i.e. no transition step occurs at all).

By Lemma [state and current symbol are unaffected by $R_\square$], the steps $D_{i_m+1} \rightarrow \dots \rightarrow D_N$ (if $m \geq 1$; or $D_0 \rightarrow \dots \rightarrow D_N$ if $m = 0$) are all $R_\square$ -steps (there is no further transition step after index i_m , by definition of i_m as the last one), so D_N has the same state q_m and current symbol c_m .

If $q_m \notin \{q_{\text{acc}}, q_{\text{rej}}\}$, then $\delta(q_m, c_m)$ is defined, so the corresponding transition rule matches D_N at the root (its left-hand side only constrains the state and current symbol, which agree), contradicting irreducibility of D_N . Hence $q_m \in \{q_{\text{acc}}, q_{\text{rej}}\}$, i.e. D_N is a halting configuration.

By lemma transition subsequence, D_{i_m+1} has the same state and current symbol as C_m from definition computation of M (or $D_0 = C_0$ if $m = 0$); since state alone determines whether a configuration is halting, C_m is also a halting configuration. Hence M halts on w (in m steps). $\square$

Lemma (*reduction of TRS termination to TM halting*). Let M be a deterministic Turing machine, and let R_M be the associated rewriting system (depending only on M , not on any particular input). Then R_M is terminating (i.e. every reduction sequence, from every term, is finite) if and only if M halts from every configuration.

Proof. ($\Rightarrow$) Assume R_M is terminating. Let C be any configuration of M . In particular, every reduction sequence issued from C is finite, hence reaches an irreducible term (R_M is finitely branching: at most one rule matches a configuration term at the root, by determinism of δ , and $R_\square$ contributes only finitely many disjoint redexes at any given term). By construction of Turing machine by a TRS, C reducing to an irreducible term means M halts from C . Since C was arbitrary, M halts from every configuration.

($\Leftarrow$) Assume M halts from every configuration. Let t be any term of R_M . If t is a configuration term C , then M halts from C , so by construction of Turing machine by a TRS, C reduces, in R_M , to an irreducible term: every reduction sequence from C is finite. Since $R_\square$ alone is terminating, and every R_M -step either progresses a configuration term (finitely many, by the above) or normalises the tape (finitely many, by termination of $R_\square$), every reduction sequence of R_M , from any term, is finite. Hence R_M is terminating. $\square$

Definition (*encoding of Turing machines*). We fix an effective encoding $\langle M \rangle \in \{0, 1\}^*$ of any deterministic Turing machine M , and identify the notion of *algorithm* with that of a Turing machine over the alphabet $\{0, 1\}$ which always halts.

Theorem (*Halting problem for Turing machines*). There is no algorithm that, given an arbitrary deterministic Turing machine M , always terminates and correctly decides whether M halts on the input $\langle M \rangle$ (its own description).

Proof. Assume, for contradiction, that such an algorithm H exists. That is, H takes as input a description $\langle M \rangle$, always halts, and outputs:

$$H(\langle M \rangle) = \begin{cases} \text{"yes"} & \text{if } M \text{ halts on } \langle M \rangle, \\ \text{"no"} & \text{if } M \text{ does not halt on } \langle M \rangle. \end{cases}$$

Using H , define a new algorithm D as follows: on input $\langle M \rangle$, D first computes $H(\langle M \rangle)$ (this always terminates, since H always halts), and then:

- if $H(\langle M \rangle) = \text{"yes"}$, D enters an infinite loop;
- if $H(\langle M \rangle) = \text{"no"}$, D halts.

D is itself an algorithm (a deterministic Turing machine), hence it has an encoding $\langle D \rangle$. Consider running D on input $\langle D \rangle$. By definition of H , exactly one of the following two cases holds.

Case 1: D halts on $\langle D \rangle$. By correctness of H , this means $H(\langle D \rangle) = \text{"yes"}$. But by definition of D , $H(\langle D \rangle) = \text{"yes"}$ forces D to enter an infinite loop on $\langle D \rangle$, i.e. D does *not* halt on $\langle D \rangle$. This contradicts the assumption of Case 1.

Case 2: D does not halt on $\langle D \rangle$. By correctness of H , this means $H(\langle D \rangle) = \text{"no"}$. But by definition of D , $H(\langle D \rangle) = \text{"no"}$ forces D to halt on $\langle D \rangle$. This contradicts the assumption of Case 2.

In both cases we reach a contradiction. Since D on input $\langle D \rangle$ must either halt or not halt (there is no third possibility), one of the two cases must occur, and each leads to a contradiction. Hence the assumption that H exists is false.

□

Theorem (*Halting problem for TRS*). There is no algorithm that, given an arbitrary term rewriting system R , always terminates and correctly decides whether R is terminating.

Proof. Assume, for contradiction, that such an algorithm A exists. Let M be an arbitrary deterministic Turing machine, and build R_M by the construction of Turing machine by a TRS. Running A on R_M decides whether R_M is terminating, hence, by the reduction lemma, decides whether M halts from every configuration. Since M was arbitrary, this contradicts the Halting problem for Turing machines. Hence no such algorithm A exists.

□